# Trenitalia
# ETR1000 VZI-R1

# Bogies

| Revision | Author | Date | Revision Description |
|---|---|---|---|
| 00 | R. Schiavo | 16/04/2024 | Generated from DOORS module /Platform.-.Zefiro-Europe/01.-.Requirements Breakdown Structure.-.Zefiro-Europe/2F.-.Vehicle-Functional Integration Areas.-.Zefiro-Europe/2F_03.-.Bogies-VF.-.Zefiro-Europe/Bogies-VF.CONCEPT, current 5.0<br><br>Doc 3ECP413719-0300_<br><br>First Emission |
| 1.1 | R. Schiavo | 18/04/2024 | Modified Hot Box Detection function according to the FA010024085 rev 01 and FA010024086 rev01 |
| 1.0 | R. Schiavo | 29/04/2024 | Emission – BL1 baseline |
| 2.1 | R. Schiavo | 08/08/2024 | Modified HABD lamp picture<br>Modified Requirements:<br>.388,.364,.380,.344,.HR25,.HR26 |
| 2.2 | R. Schiavo | 29/08/2024 | New requirement<br>.HR28<br>Modified<br>.HR20, .HR21 |
| 2.3 | R. Schiavo | 03/09/2024 | TB Desk lamp lighting if overtemperature alarms occurr |
| 2.4 | R. Schiavo | 04/09/2024 | gear and axlebox overtemperature alarms classified as Warning. Modified requirements: .Hr02,.HR03,.364,.117,.HR18,.HR16<br>gear and axlebox overtemperature alarms classified as C2 serious Fault. Modifeied Requirements: .388,.HR30,.377,.362<br>Deleted requirement .HR21 because it is present in the CTC FRS requirements (FA020023107)<br>Modified HABD C1/C2 reset. Modified requirement .380, .HR28,. HR29 |
| 2.00 | R. Schiavo | 16/09/2024 | Emission BL2 baseline |
| 3.1 | R. Schiavo | 09/01/2025 | Deleted req .HR20 because moved in CTC FRS FA020023107 |
| 3.2 | R. Schiavo | 21/01/2025 | Modified case 5 in the table1 adding condition reported in the requirement .364<br>Modified requirements<br>.371,.388,.HR04, HR19, .117, .<br>HR17, .HR18, .359, .241, .HR08, .<br>HR09, .HR10,.HR11,.HR12, .HR13<br>Added requirements:<br>.HR31,.HR32,.HR33,.HR34 |

| | | | |
|---|---|---|---|
| 3.00 | R. Schiavo | 21/01/2025 | Emission BL3 baseline |
| 4.01 | G.Spina | 11/06/2025 | Modified requirements:<br>TRS.56,<br>HR01, HR30, HR04, HR16, HR25<br>CONCEPT.368, 392, 254, 364, 141, 377, 344 |
| 4.0 | G.Spina | 06/08/2025 | GPLM release BL4 |

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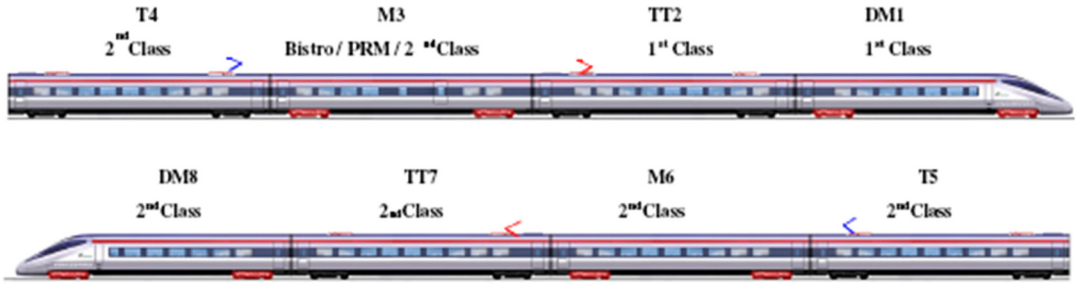
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# 1Introduction

This document is a function specification for the control of the *XXX* system for V300ZEFIRO for customer Trenitalia.

## Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:




- DM= End driving motor car with driver's cab
- TT= Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

## Definitions and abbreviations

CDCircuit Diagram
EVENTDiagnostics that will be stored in the Event database within TCMS SW.
HW HMIHardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDUInterface Display Unit
TCMSTrain Control and Management System
IMSInstability Monitoring System
ALSActive Lateral Suspension
CBMCondition Based Maintenance
WFLWheel flange lubrication
HABDhot axlebox detection
SPMShock-Pulse Method
HABDHot axle box detection
DNRADetection of non-rotating axle
CANController Area Network
TCMSTrain Control Management System
TRSTechnical requirements specification
SFRSSystem Function Requirement Specification
ICDInterface Control Document
IPInternet Protocol
GPSGlobal Positioning System
FVDSFunctional Vehicle Design Specification

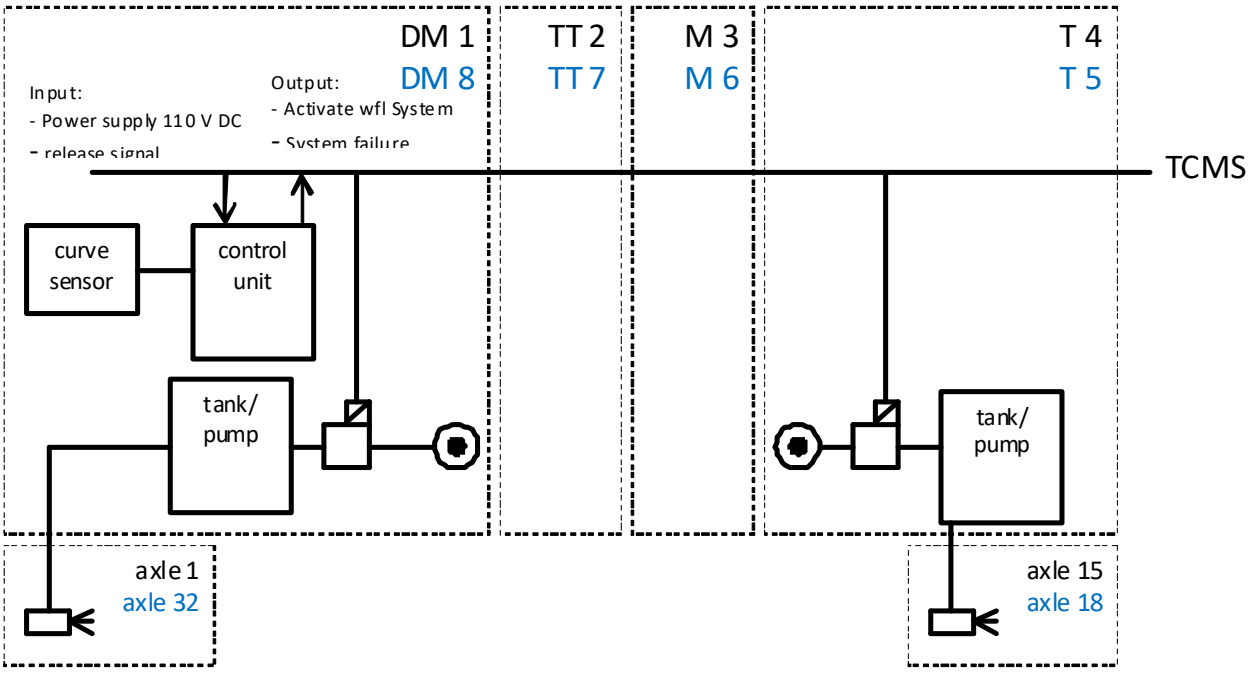
# 2 System Overview

This SFRS covers the flange lubrication, hotbox detection, instability monitoring system, active lateral suspension and diagnostics & condition based maintenance. A separate overview is given for every system.
The Bogie Controller manages the ALS function of both bogies of the same car to improve the comfort levels.
The CBM Controller monitors the conditions of certain bogie components to optimise the maintenance interventions by detecting degradations.

## System Configuration

### Flange Lubrication

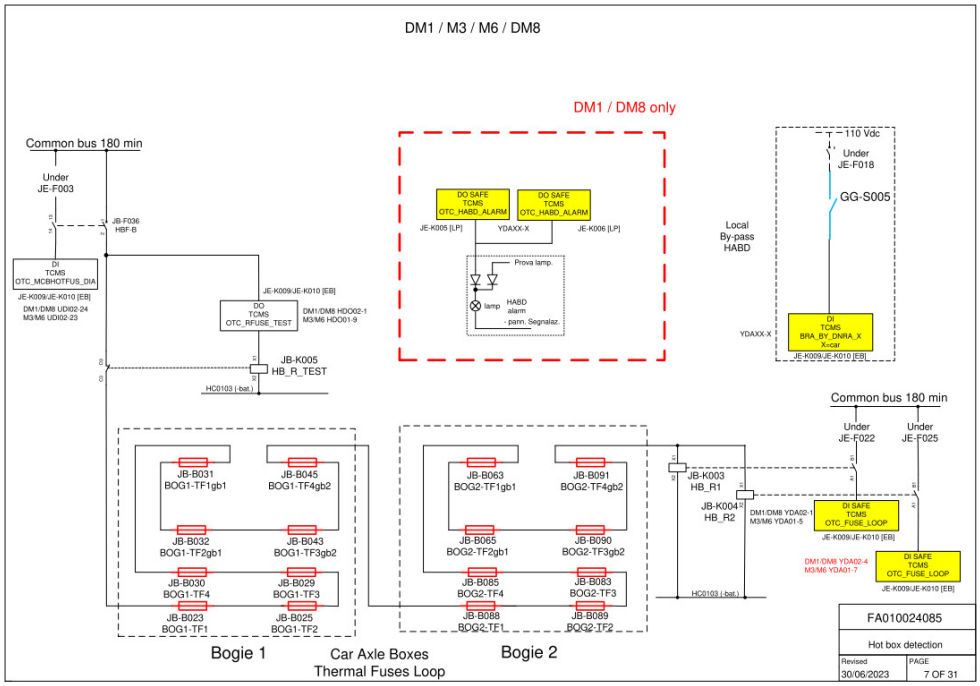
WFL schematic



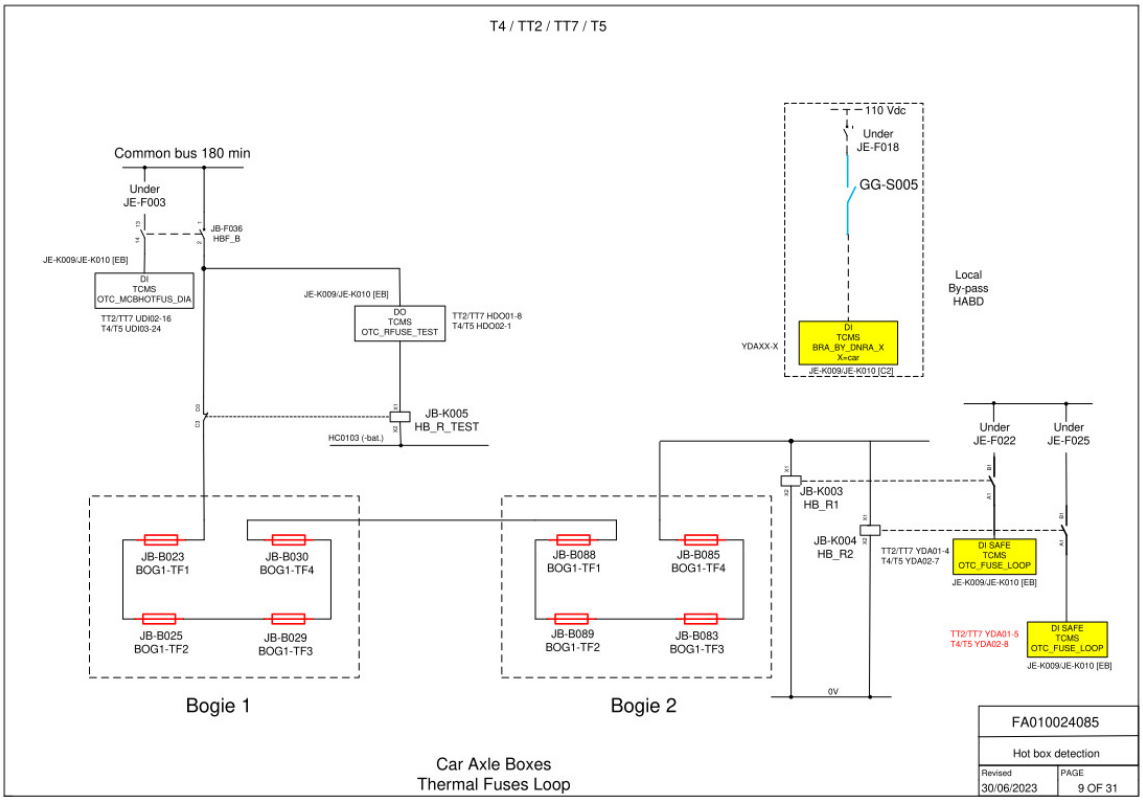

The wheel flange lubrication system consists of two separate and independent units, each for one running direction. Each unit consists of one control unit including curve sensor and two lubrication units. One for axle 1 and one for axle 15 and accordingly axle 32 and axle 18.

## Hotbox Detection

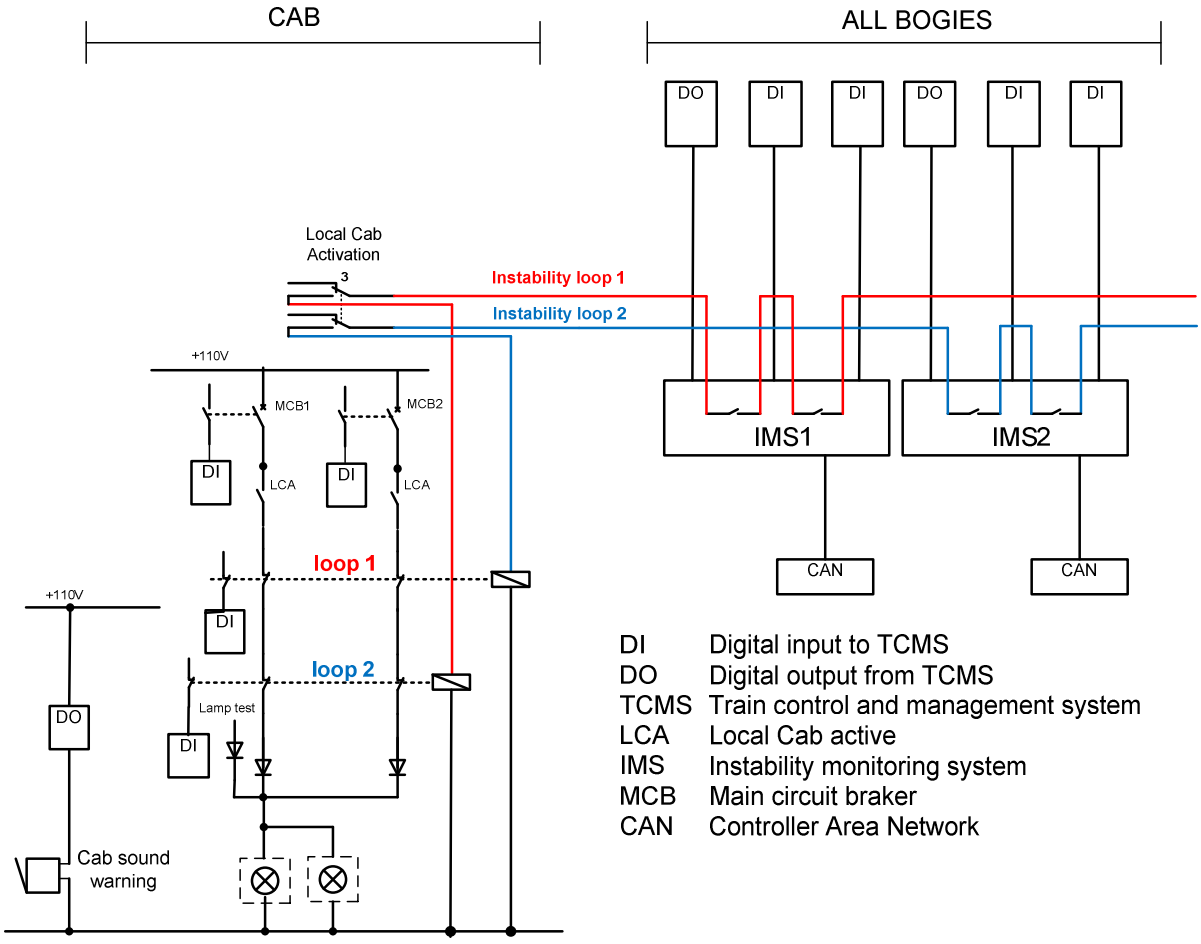
Hot axlebox detection – safe loop motor car:




Hot axlebox detection – safe loop trailer car:



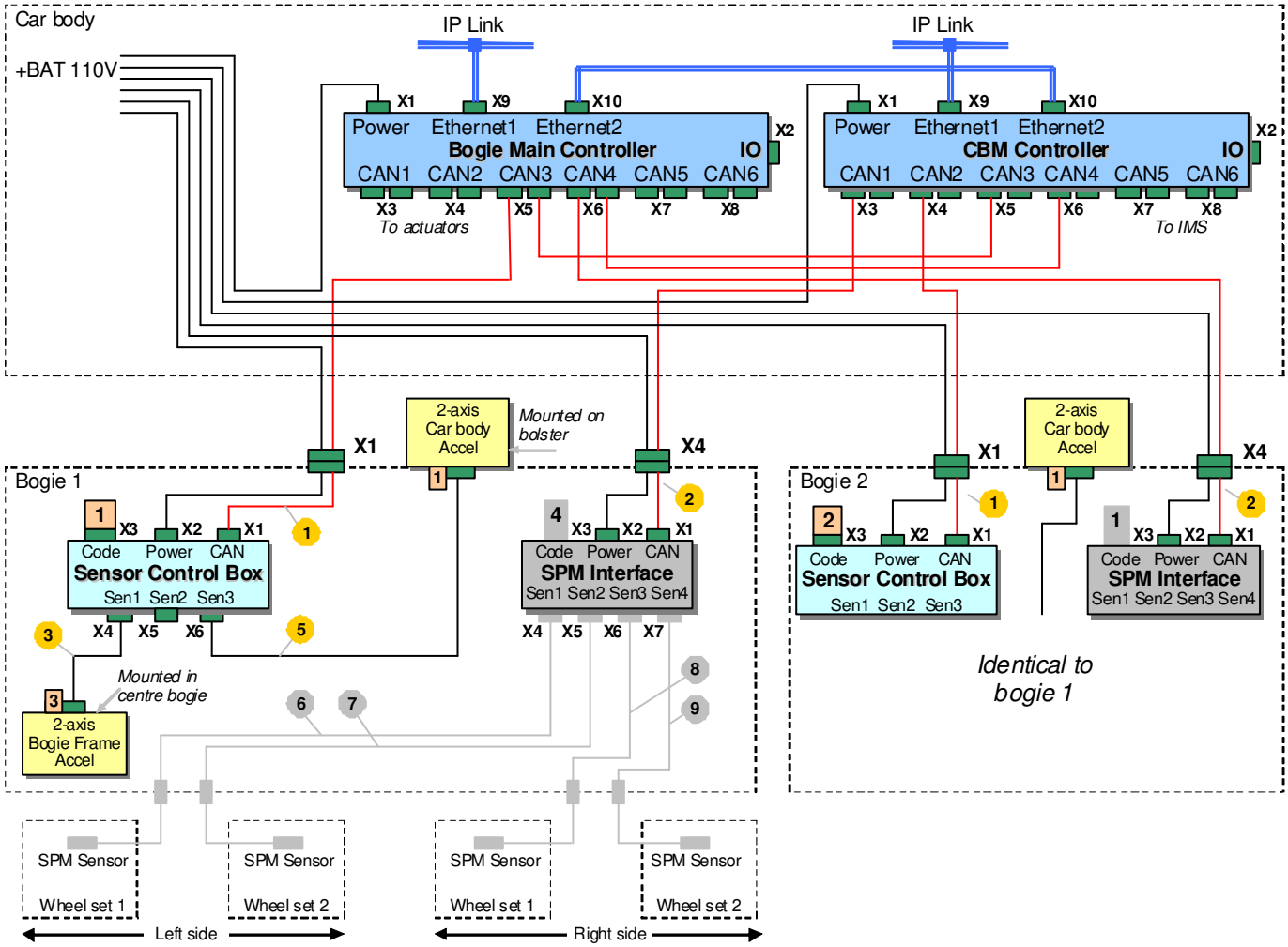
## Instability Monitoring




| | |
|---|---|
| DI | Digital input to TCMS |
| DO | Digital output from TCMS |
| TCMS | Train control and management system |
| LCA | Local Cab active |
| IMS | Instability monitoring system |
| MCB | Main circuit braker |
| CAN | Controller Area Network |

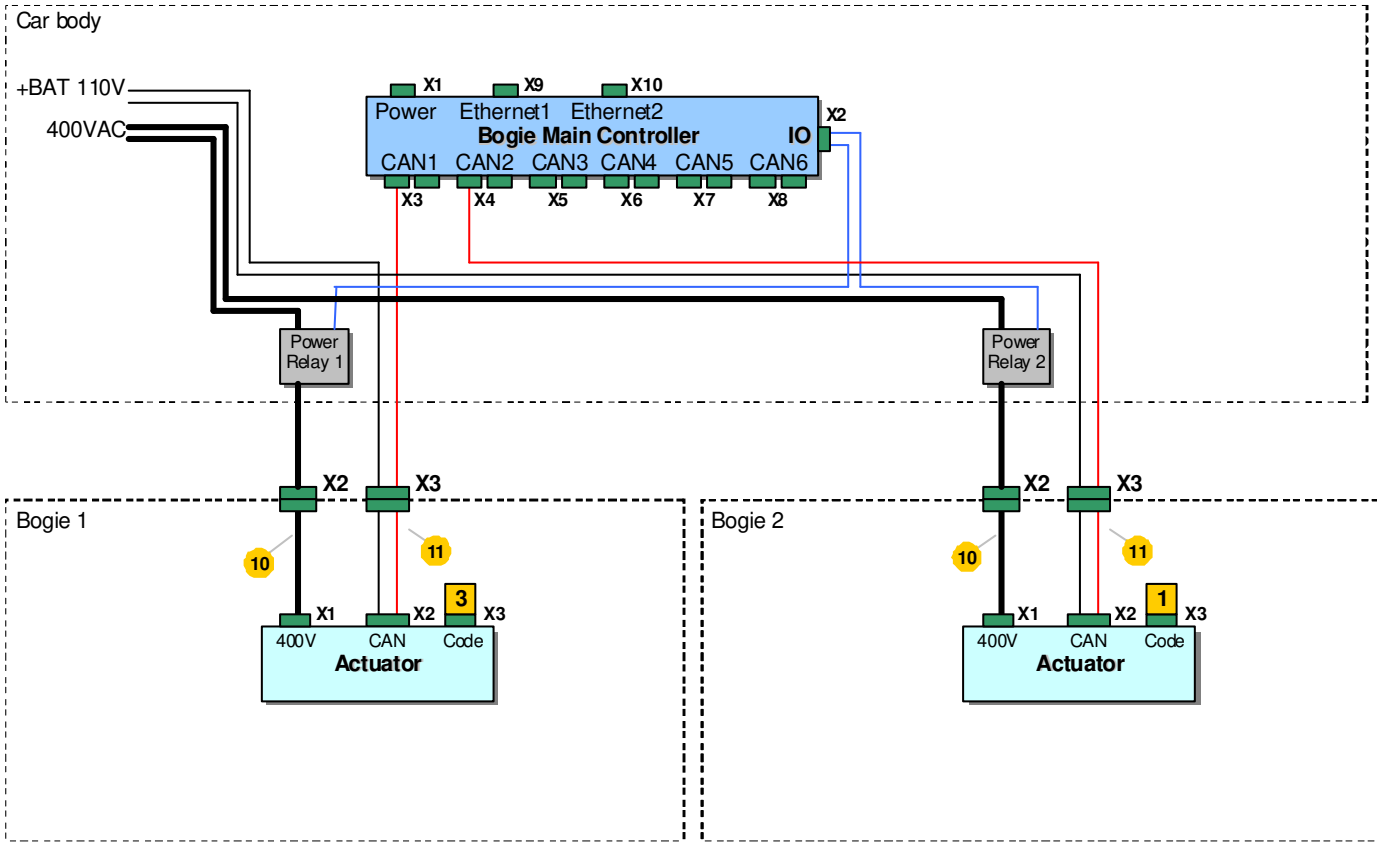
## Active Lateral Suspension

**Sensor configuration: ALS-CBM**




Note: the SPM interface and the SPM sensor are considered as an option

**Actuator circuit: ALS**



# 3Functional requirements

## Flange Lubrication

### Function Design

It is the purpose of the wheel flange lubrication system to reduce the wear between wheel flange and rail by using specialized lubricants. The lubricant is applied by means of air as a fine spray onto the wheel flange via single medium spray nozzles. As described in 2.1.1. the WFL consists of two symmetric units. Only one unit is described in this part. All requirements are accordingly valid for the second unit. The amount of lubricant per spray cycle that is applied on the wheel flange and activation of a spray cycle is controlled by the WFL control unit.

### Normal condition

If all requirements on the train side are fulfilled, the flange lubrication is active. An active WFL system releases a spray signal if the integrated curve sensor. This signal is linked to the solenoid valves via TCMS. They operate when the signal is applied and a spray of grease is released.

#### *Failure cases*

1. Lubrication applied when not ordered
2. Lubrication not applied when ordered
3. Failure of flange lubrication control unit

#### *Failure detection*

1. No detection of failure during service possible. Detection of failure during normal maintenance by monitoring the amount of lubricant that is refilled.
2. No detection of failure during service possible. Failure detected during maintenance by use of integrated test function.
3. Event of WFL failure shall be generated and stored in TCMS

#### *Failure handling*

1. System repaired when failure detected during scheduled maintenance.
2. System repaired when failure detected during scheduled maintenance.
3. System shall be repaired during next scheduled maintenance

#### *Failure effect on operation*

1. No effect
2. No effect
3. No effect

### Requirements

| | |
|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.14 |
| **Description** | Wheel flange lubrication shall be installed on at least two axle of the leading bogie. It shall operate curve dependently. |
| **Safety level** | N/A |

One WFL unit consists of one control unit(inc. curve sensor) located in the cab and two lubrication units located in the underframe in front of axle 1 and axle 15. The tanks should be installed where quick refill by means of a filler coupling and the use of a separate filler pump or after opening the filler lid is possible.
The length of the tubing system from the tank to the nozzles shall be as short as possible.
The control unit installed in the cab must be easily accessible

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.167 | A 'spray impulse' signal shall be generated when a curve is detected by the WFL system. | Derived Requirement | 4S_03.06.02.-.Flange Lubrication Systems | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.156 | The 'spray impulse' signal shall be linked from the control unit to TCMS. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control – Circuits | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.279 | The solinoid valves on the lubrication units on axle 1 in the DM 1 car and axle 15 in the T4 car shall be linked to TCMS | Derived Requirement | 4S_09.02.09.-.Conventional Train Control – Circuits | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.280 | If the 'spray impulse' signal is set by the control unit, both solenoid valves of the active wfl system shall be activated. | Derived Requirement | 4S_09.03.05.-.TCMS Software | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.325 | The control unit installed in the cab must be easily accessible | Derived Requirement | 4S_08.07.01.-.Drivers Desk | | N/A |

| ID | 2F_03.Zefiro-Europe.TRS.11 |
|---|---|
| Description | The lubrication shall be enabled if<br>– speed v > 3 km/h<br>– sanding not active<br>– train speed v <= 160km/h |
| Safety level | N/A |

The input requirement specifies the case when the wheel flange lubrication system must be enabled. This is derived to TCMS Software to generate an 'enable wheel flange lubrication' signal. Therefore only one signal has to be send to the lubrication control unit. Which of the control units shall receive the 'enable wheel flange lubrication' signal depends on the running direction. It always has to be the control unit in the leading cab of the train. The cab activation signal can't be used for this purpose because in this case the wheel flanges of a second train in multi traction would not be lubricated. That means, that the 'enable wheel flange lubrication' is only valid on train level.

Additionally to the enabling signal a fault signal must be added to generate an event in the diagnostic, when a detectable fault in the wheel flange lubrication unit occurs.

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.108 | The output of TCMS 'enable wheel flange lubrication' shall be set for the control unit in the leading cab of the train when<br>– speed v > 3 km/h<br>– sanding not active<br>– train speed v <= 160km/h | Derived Requirement | 4S_09.03.05.-.TCMS Software | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.350 | The 'enable wheel flange lubrication' signal shall to be linked from TCMS to the Flange lubrication control box. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control – Circuits | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.157 | If the wheel flange lubrication control unit input 'enable wheel flange lubrication' is set the wheel flange lubrication system shall be enabled | Derived Requirement | 4S_03.06.02.-.Flange Lubrication Systems | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.158 | If the output of the wheel flange lubrication control unit 'wheel flange lubrication fault' is set to 0 for >2 seconds an event shall be set in the diagnostic system.<br>If the output is set to 1 the system shall be considered as Ok. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.351 | The 'wheel flange lubrication fault' signal shall to be linked from the Flange lubrication control box to TCMS | Derived Requirement | 4S_09.02.09.-.Conventional Train Control – Circuits | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.159 | The output of the wheel flange lubrication control unit 'wheel flange lubrication fault' shall be set to 0 when any kind of failure is detected by the control unit. In case of no failure the signal shall be set to 1. | Derived Requirement | 4S_03.06.02.-.Flange Lubrication Systems | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.282 | The description of an © failure shall be displayed on the control unit | Derived Requirement | 4S_03.06.02.-.Flange Lubrication Systems | | N/A |

| ID | 2F_03.Zefiro-Europe.TRS.16 |
|---|---|
| Description | Spraying of the grease shall be triggered by a pulse from the lateral accelerometer if a curve is detected. The wheel/rail contact area shall not be contaminated. |
| Safety level | N/A |

The grease of the wheel flange lubrication system must be sprayed when the limit value of the curve sensor is reached for more than a specific period of time to avoid a spray to be triggered by other short-time exceedings of the limit value.

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.111 | The spraying of the grease shall be triggered when the curve signal of the curve sensor is set for >=2s(tbd) | Derived Requirement | 4S_03.06.02.-.Flange Lubrication Systems | | N/A |

| | | | | | |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.278 | The detection level of the curve sensor shall be adaptable | Derived Requirement | 4S_03.06.02.-.Flange Lubrication Systems | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.161 | The spraying of the grease shall be triggered in a defineable periode | Derived Requirement | 4S_03.06.02.-.Flange Lubrication Systems | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.162 | When triggered the grease shall be sprayed for a definable periode of time | Derived Requirement | 4S_03.06.02.-.Flange Lubrication Systems | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.163 | The wheel/rail contact area shall not be contaminated | Derived Requirement | 4S_03.06.02.-.Flange Lubrication Systems | | N/A |

| | |
|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.20 |
| **Description** | Test of the lubrication system shall be possible in maintenance mode by using:<br>– a push button on the lubrication control unit |
| **Safety level** | N/A |

A selftest function has to be integrated in the WFL system. It should be possible to release a spray of grease in standstill to visually check the amount of grease sprayed on the wheel flange

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.114 | A button and function must be integrated in the lubrication control unit to release a spray of grease for test purpose. | Derived Requirement | 4S_03.06.02.-.Flange Lubrication Systems | | N/A |

# Hotbox Detection

## Function Design

The hot axle box detection (HABD) monitors the temperature of each wheelset bearing which indicates damage of the bearings that may lead to critical accidents. Therefore this function is highly safety relevant.
The supervision is done from TCMS by two separate onboard systems.

- Measurement of the temperature by Pt100 Sensor . If certain temperature thresholds are exceeded, activation of alarm shall be set and events shall be stored in the diagnostic system of the train.
- Detection of critical temperature threshold by thermal fuses which opens a safety loop of the car and can lead to the activation of an alarm

The melting of one of the fuses opens the car loop of the HABD system. The car loop status is acquired from TCMS by the open/closed contacts of 2 relays.The HABD alarm occurs if the car loop is interrupted and at least an axle box or gear box PT100 detects a preAlarm condition or at least a Pt100 is faulty or excluded or the measured temperature value is not plausible. The alarm occurs also if a Pt100 reaches the fuse melt threshold and the loop is not interrupted.
The HABD alarm is indicated to the driver by the activation of the HABD-lamp on the driver's desk and the main cab buzzer. The TCMS triggers also a notification on the IDU showing the alarm pictogram and provides the identification of the affected bogie
The TCMS actions, the Alarms and the diagnostic events are reported in the Table 1. The actions requested to the Driver are also reported.

### *Normal condition*

Continuous measurement of the axle box bearing temperatures. If over-temperature is detected for a bogie bearing an event must be set according to the rate of the temperature.
If the temperature of a bearing is too high, a thermal fuse melts and opens the safety car loop

### *Failure cases*

1. Overtemperature is present but not detected by Pt100
2. Overtemperature is not present but indicated by Pt100.
3. Thermal fuse melts or bogie loop is broken but no overtemperature is present
4. Thermal fuse melts, but a safety relay is welded closed.
5. Thermal fuse doesn't melt but Pt100 detects overtemperature.

### *Failure detection*

1. If a faulty pt100 is present in the car the overtemperature Alarm is generated also if only a thermal fuse melts. The check of temperature doesn't show which axle box is effected but the TCMS supervision of the bogie loop indicates which car is effected.
2. If the pt100 is not considered faulty the alarm is generated. Otherwise, an Event will be set in the diagnostic system.
3. Train does not consider the situation as a real hot box A maintenance event of undue loop interruption is generated
4. TCMS supervises both relays and if they are not aligned a diagnostic event –s set - Implausibility of safety relays
5. Event in the diagnostic system due to no detection by loop. The overtemperature alarm is generated

### *Failure handling*

1. Replacement of sensor during repair. Repair of axle box/bearing during unplanned workshop visit
2. Replacement of sensor and check of axle box during next scheduled maintenance.
3. Check/repair of all bogie loop parts during unplanned workshop visit.
4. Repair of axle box/bearing during unplanned workshop visit.
   Replacement/check of safety relays in the unplanned workshop visit.
5. Axle box/bearing checked/repaired during next scheduled maintenance.

### *Failure effects on operation*

1. Train is stopped due to the alarm.
2. If alarm is generated the train is stopped. If the pt100 is considered faulty No effect
3. No effect if undue broken loop.
4. Train is stopped due the alarm
5. Train is stopped due the alarm

| Case n. | Thermal Fuses loop coach w | Pt100 coach w, bearing x | Pt100 coach w, gear axle y | Pt100 Faulty coach w | HABD Alarm & Driver Event | Diagnostic Events | TCMS action | Driver Guide | Effect on the Operation |
|---|---|---|---|---|---|---|---|---|---|
| 1 | x | - | - | - | "Undue HABD car loop interruption" warning | thermal fuses loop undue interrupted in coach | - | - | - |
| 2 | Not interrupted | 1)Tx<-50°C OR Tx>500°C<br>2)\|dTx/dt\|>10°C/sec<br>3)\|dTx/dt\|/Max(\|dTi/dt\|)>=4 AND Tx<0<br>4) Tx<Medium(Ti)-30°C | - | x | - | 1)Bearing x–Pt100 - faulty<br>2)Bearing x Pt100 – excluded (A)<br>3)Bearing x Pt100 – ex©ded (C)<br>4) Bearing x Temperature value is not plausible | - | - | - |
| 3 | Not interrupted | | 1)5)Ty<-50°C OR Ty>500°C<br>2)6)\|dTy/dt\|>10°C/sec<br>3)7)\|dTy/dt\|/Max(\|dTi/dt\|)>=4 AND Ty<0<br>4)8) Ty<Medium(Ti)-30°C | x | - | 1)Gear axle y Pt100 –ast side - faulty<br>2)Gear axle y Pt100 fast side – excluded (A)<br>3)Gear axle y Pt100 fast side –©cluded (C)<br>4)Gear axle y Pt100 fast side – value Not plausible<br>5)Gear axle y Pt1–0 slow side - faulty<br>6)Gear axle y Pt100 slow side – excluded (A)<br>7)Gear axle y Pt1–0 slow sid© excluded (C)<br>8)Gear axle y Pt100 slow side – value Not plausible | - | - | - |
| 4 | Not interrupted | Tx>95°C for more than 30 | - | - | *TOD pictogram:*<br>  <br>*Axle box overtemperature pre-Alarm detected by PT100 Warning* | Axle Box Temperature PreAlarm coach w, bearing x | - | A speed reduction to 200 km/h is requested to Driver | Speed limitation to 200km/h |
| 5 | Not interrupted | - | Ty> 135°C for more than 30 s<br>OR<br>gradient >7.5°C in 30s where the starting temperature is >80°C<br>OR<br>Motor side bearings (fast side) sensor difference | - | *TOD Pictogram:*<br>  <br>*Gear box overtemperature pre-Alarm detected by PT100 Warning* | Gear Temperature PreAlarm coach w, axle y | - | A speed reduction to 200 km/h is requested to Driver | Speed limitation to 200km/h |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | \|T1-T2\|>30°C in 30 s, OR. Wheelset side bearing (slow side) sensor difference \|T1-T2\|>25°C in 30 sec | | | | | | |
| **6** | Not interrupted | Tx > 110°C for more than 30sec | - | - | *TOD pictogram:*  *Driver Desk lamp*  *Axle box overtemperature Alarm detected by PT100 Serious Fault C2* | Axle Box overTemperature Alarm coach w, bearing x | Traction cutout if speed >80km/h<br>Vehicle stopping requested to Driver as soon as possible | At standstill::<br>Check bearings temperature on the coach x.<br>If Alarm confirmed free the track at low speed (20km/h)<br>If no anomaly is detected the temperature monitoring (HABD Function) is excluded in this coach by local bypass (GG-S005) and the speed is limited to 200km/h from TCMS | Vehicle stopping<br>If alarm confirmed the traini s out of service<br>If alarm not confirmed the speed is limited to 200km/h |
| **7** | Not interrupted | | Ty > 175°C<br>OR<br>- Fast side Difference between both sensors \|Ty1-Ty2\|>40°C<br>OR<br>- Slow Side Difference between both sensors \|Ty1-Ty2\|>35°C all condition for more than 30 s<br>OR<br>dTy/dt > 12,5°C/30s with starting temperature is >80°C | - | *TOD pictogram:*  *Driver Desk lamp*  *Gear box overtemperature Alarm detected by PT100 Serious Fault C2* | Gear Box overTemperature Alarm coach w, axle y | Traction cutout if speed >80km/h<br>Vehicle stopping requested to Driver as soon as possible | At standstill::<br>Check bearings temperature on the coach x.<br>If Alarm confirmed free the track at low speed (20km/h)<br>If no anomaly is detected the temperature monitoring (HABD Function) is excluded in this coach by local bypass (GG-S005) and the speed is limited to 200km/h from TCMS | Vehicle stopping<br>If alarm confirmed the traini s out of service<br>If alarm not confirmed the speed is limited to 200km/h |
| **8** | Not interrupted | Tx> 121°C for more than 30sec | - | - | *TOD pictogram:*  *Driver Desk lamp*  *Acoustic signal* *Axle Box excessive temperature HABD Alarm detected by Pt100 Serious Fault C1* | Axle Box excessiveTemperature HABD Alarm coach w, bearing x | Automatic speed reduction to 80km/h and then 80km/h speed limitation<br>Vehicle stopping requested to Driver as soon as possible | At standstill::<br>Check bearings temperature on the coach x.<br>If Alarm confirmed free the track at low speed (20km/h)<br>If no anomaly is detected the temperature monitoring (HABD Function) is excluded in this coach by local bypass (GG-S005) and the speed is limited to 200km/h from TCMS | Vehicle stopping<br>If alarm confirmed the traini s out of service<br>If alarm not confirmed the speed is limited to 200km/h |
| **9** | Not interrupted | - | Ty>184°C for more than 30sec | - | *TOD pictogram:*  *Driver Desk lamp*  *Acoustic signal* *Gear Box excessive temperature HABD Alarm detected by Pt100 Serious Fault C1* | Gear Box excessive Temperature HABD Alarm coach w, axle y | Automatic speed reduction to 80km/h and then 80km/h speed limitation<br>Vehicle stopping requested to Driver as soon as possible | At standstill::<br>Check bearings temperature on the coach x.<br>If Alarm confirmed free the track at low speed (20km/h)<br>If no anomaly is detected the temperature monitoring (HABD Function) is excluded in this coach by local bypass (GG-S005) and the speed is limited to 200km/h from TCMS | Vehicle stopping<br>If alarm confirmed the traini s out of service<br>If alarm not confirmed the speed is limited to 200km/h |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **10** | Interrupted | Temperature condition case 4or 6 | | | *TOD pictogram*<br>  <br>*Driver Desk lamp*<br>  <br>*Acoustic signal*<br>*Axle Box excessive temperature HABD Alarm detected by Thermal Fuses*<br>*Serious Fault C1* | Axle Box excessiveTemperature HABD Alarm coach w, bearing x | Automatic speed reduction to 80km/h and then 80km/h speed limitation<br>Vehicle stopping requested to Driver as soon as possible | At standstill::<br>Check bearings temperature on the coach x.<br>If Alarm confirmed free the track at low speed (20km/h)<br>If no anomaly is detected the temperature monitoring (HABD Function) is excluded in this coach by local bypass (GG-S005) and the speed is limited to 200km/h from TCMS | Vehicle stopping<br>If alarm confirmed the traini s out of service<br>If alarm not confirmed the speed is limited to 200km/h |
| **11** | Interrupted | - | Temperature condition case 5 or 7 | - | *TOD pictogram:*<br>  <br>*Driver Desk lamp*<br>  <br>*Acoustic signal*<br>*Gear Box excessive temperature Alarm detected by Thermal Fuses*<br>*Serious Fault C1* | Gear Box excessive Temperature HABD Alarm coach w, axle y | Automatic speed reduction to 80km/h and then 80km/h speed limitation<br>Vehicle stopping requested to Driver as soon as possible | At standstill::<br>Check gear temperature on the coach x.<br>If Alarm confirmed free the track at low speed (20km/h)<br>If no anomaly is detected the temperature monitoring (HABD Function) is excluded in this coach by local bypass (GG-S005) and the speed is limited to 200km/h from TCMS | Vehicle stopping<br>If alarm confirmed the traini s out of service<br>If alarm not confirmed the speed is limited to 200km/h |
| **12** | Interrupted | Pt100 condition case 2 | | x | *TOD pictogram:*<br>  <br>*Driver Desk lamp*<br>  <br>*Acoustic signal*<br>*Gear Box excessive temperature HABD Alarm detected by Thermal Fuses*<br>*Serious Fault C1* | Gear Box excessive Temperature HABD Alarm coach w, axle y<br>One of these events<br>1)Bearing x Pt100 - faulty<br>2)Bearing x Pt100 – excluded (A)<br>3)Bearing x Pt100 – excluded (C)<br>4) Bearing x Temperature value is not plausible | Automatic speed reduction to 80km/h and then 80km/h speed limitation<br>Vehicle stopping requested to Driver as soon as possible | At standstill::<br>Check bearings temperature on the coach x.<br>If Alarm confirmed free the track at low speed (20km/h)<br>If no anomaly is detected the temperature monitoring (HABD Function) is excluded in this coach by local bypass (GG-S005) and the speed is limited to 200km/h from TCMS | Vehicle stopping<br>If alarm confirmed the traini s out of service<br>If alarm not confirmed the speed is limited to 200km/h |
| **13** | Interrupted | - | Pt100 condition case 3 | x | *TOD pictogram:*<br>  <br>*Driver Desk lamp*<br>  <br>*Acoustic signal*<br>*Gear Box excessive temperature HABD Alarm detected by Thermal Fuses*<br>*Serious Fault C1* | Gear Box excessive Temperature HABD Alarm coach w, axle y<br>One of these events:<br>1)Gear axle y Pt100 fast side - faulty<br>2)Gear axle y Pt100 fast side – excluded (A)<br>3)Gear axle y Pt100 fast side – excluded (C)<br>4)Gear axle y Pt100 fast side – value Not plausible<br>5)Gear axle y Pt100 slow side - faulty<br>6)Gear axle y Pt100 slow side – excluded (A)<br>7)Gear axle y Pt100 slow side - excluded (C)<br>8)Gear axle y Pt100 slow side – value Not plausible | Automatic speed reduction to 80km/h and then 80km/h speed limitation<br>Vehicle stopping requested to Driver as soon as possible | At standstill::<br>Check gear temperature on the coach x.<br>If Alarm confirmed free the track at low speed (20km/h)<br>If no anomaly is detected the temperature monitoring (HABD Function) is excluded in this coach by local bypass (GG-S005) and the speed is limited to 200km/h from TCMS | Vehicle stopping<br>If alarm confirmed the traini s out of service<br>If alarm not confirmed the speed is limited to 200km/h |

**Table 1**

The following pictures report the fuses car loop and the PT100 positions:

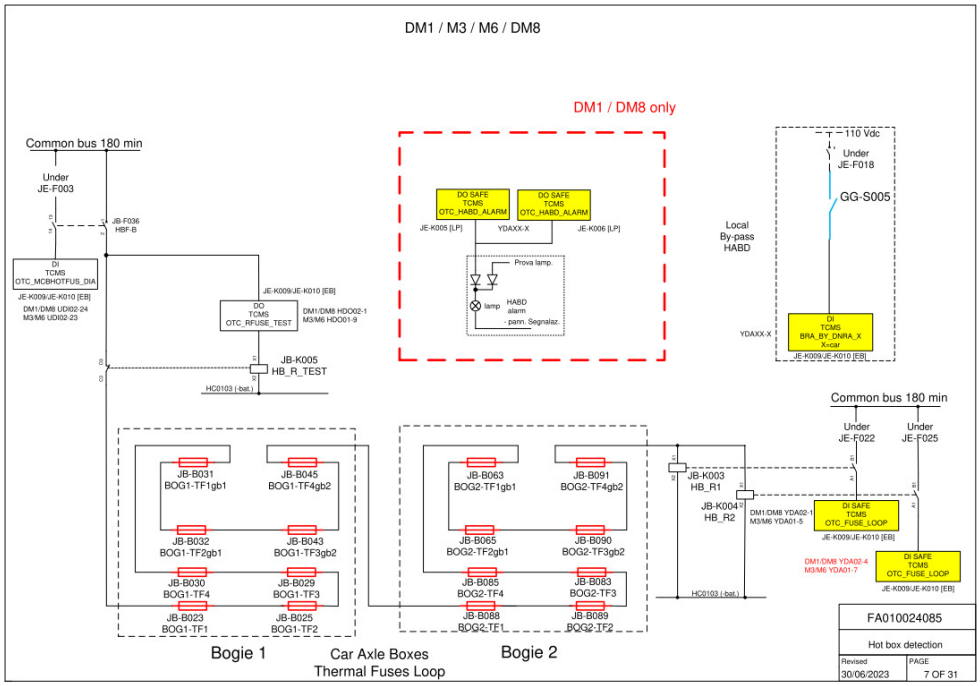



**Figure 1 Circuit diagram of HABD motor car loop**

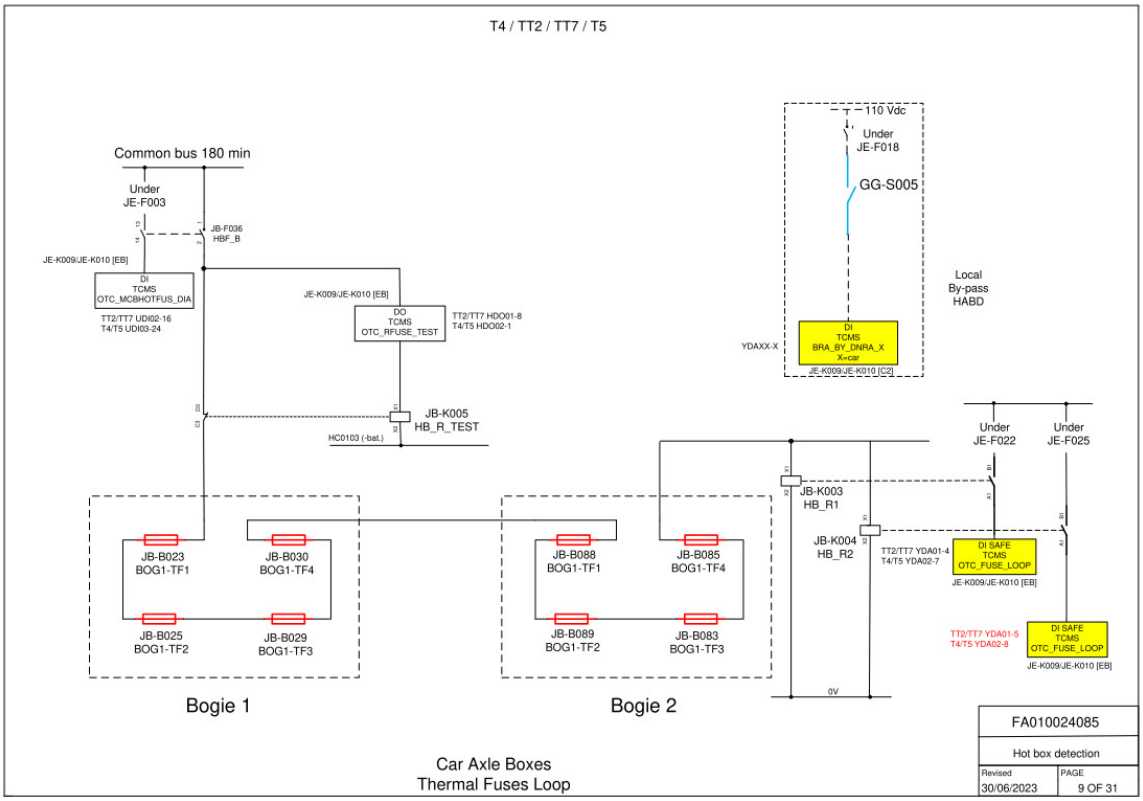



**Figure 2 Circuit diagram of HABD trailer car loop**

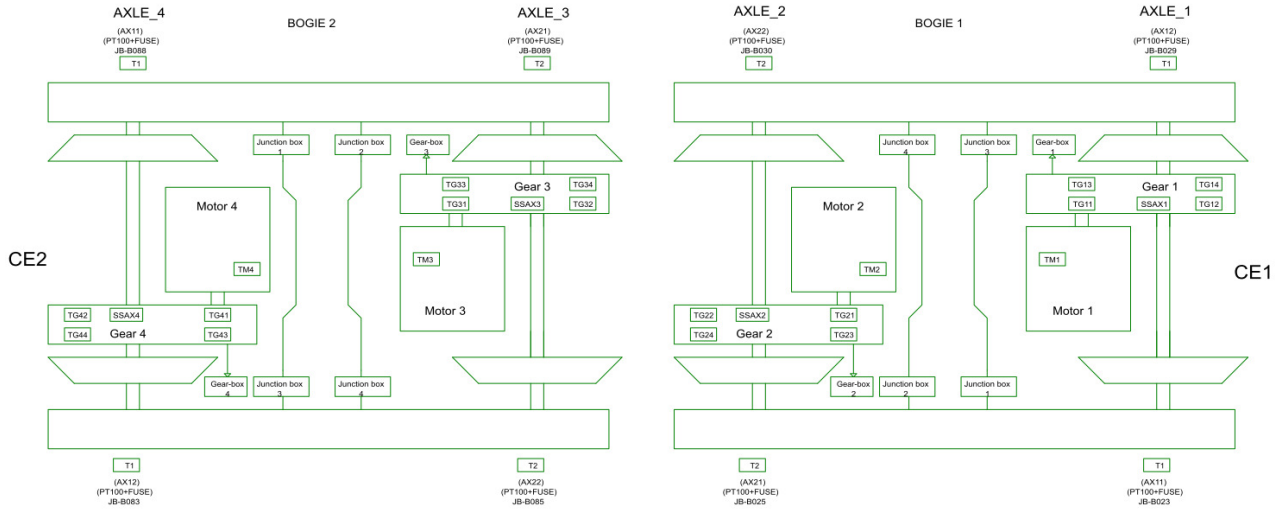



**Figure 3 Motor car PT100 and thermal fuses**

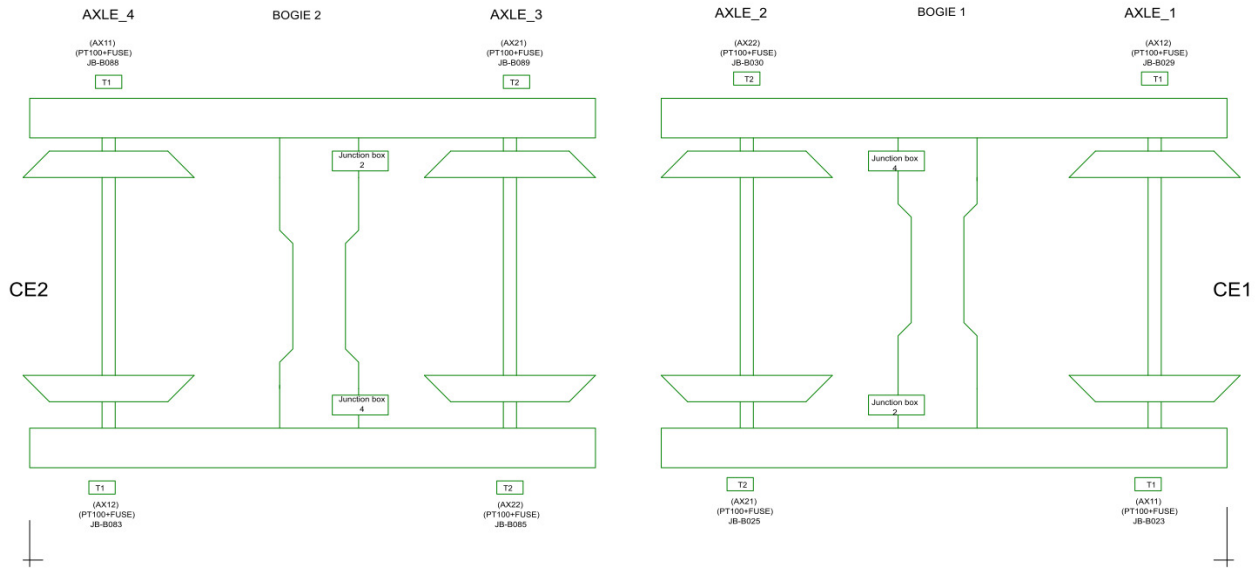



**Figure 4 Trailer car PT100 and thermal fuses**

## Requirements

### *Hot axle box detection safety loop*

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| **ID** | 2F_03.Zefiro-Europe.TRS.67 |
| **Description** | The temperature of all axle bearings shall be supervised by hot box detectors installed on board the train. |
| **Safety level** | SIL 2 |

| | |
|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.56 |
| **Description** | If any hotbox detector reports "over temperature" then the driver shall be notified(also in multiple composition). |
| **Safety level** | SIL 2 |

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.255 | All thermal fuses of a bogie must be connected in serial to a car safety loop which is deenergized when one thermal fuse melts. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.349 | A thermal fuse shall be installed in every axle box. | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.256 | The car loop shall be connected in parallel to two power saftey relais (HB_R1, HB_R2) | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.367 | Activating a sealed switch, it shall be possible to bypass HABD function separately in each coach. The bypass switch status is connected to TCMS safe Digital inputs | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.HR01 | Activating a sealed switch in the trainset, it shall be possible to disable the local hot box detection<br>Switches-off HABD lamp and pictogram on the TOD if activated for the alarm detected in the car where the bypass is activated,<br>The HABD function remains active on the other cars | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |

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| 2F_03.Zefiro-Europe.CONCEPT.368 | In case HABD bypass car switch is activated in the trainset, an event shall be set (warning to the driver) | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.392 | If any HABD bypass car switch is activated in the trainset:<br><br>- speed limitation shall be set to the driver reporting 200km/h. | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.318 | The contact of the safety relais (HB_R1, HB_R2) are connected to TCMS safe Digital input | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 2 |
| 2F_03.Zefiro-+Europe.CONCEPT.HR06 | The thermal fuses loop is considered interrupted if at least one of relay (HB_R1, HB_R2) is de-enrgized | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.257 | The hot axle box detection is performed from TCMS by safety relais and the Pt100 sensors (8 sensors from axle box sensors per each car and 8 sensors from gearbox sensors per bogie only in the motor cars). | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |

*Temperature supervision*

| | |
|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.102 |
| **Description** | If any hotbox detector reports "over temperature" then an event shall be set in the diagnostic system. |
| **Safety level** | N/A |

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|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.109 |
| **Description** | If any gearbox sensor detects "over temperature" then an event shall be set in the diagnostic system and the driver shall be notified |
| **Safety level** | |

The temperature detection shall have two thresholds to distinguish between two different damage states of the bearings according to EN 15437-2.

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.60 | Pt100 Sensor shall be installed in every wheel axle box. | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.HR06 | A HABD alarm lamp connected to TCMS safe output shall be provided on driver's desk in the cab. The HABD lamp shall be visible to driver | Derived Requirement | 4S_08.07.01.-.Drivers Desk | HW HMI | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.271 | The Pt100 sensors shall be connected to safe TCMS interfaces, 4 wire technology must be used to measure the temperature. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.254 | Temperature signals and axle box identification shall be displayed on IDU. All gearbox temperatures shall displayed on the IDU(also in multiple composition). | Derived Requirement | 4S_09.03.05.-.TCMS Software | IDU | SIL 0 |

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| 2F_03.Zefiro-Europe.CONCEPT.HR02 | **IF**<br>A Gearbox temperature >= 135 °C for more than 30 s after a low pass filter with T=8sec<br>**AND**<br>Thermal fuse loop in the same car is not interrupted<br>**THEN**<br>"Gearbox overTemperature preAlarm detected by PT100" driver Warning is generated<br>Each event shall report the information about which car and axle is affected<br>The Gearbox PreAlarm Pictogram on the TOD shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.HR03 | TCMS shall monitor single bearing of each gearbox and set following events based on the gradient evaluated over 30 sec.<br>**IF**<br>– PreAlarm: gradient > 7,5°C where the starting temperature is >80°C<br>**AND**<br>Thermal fuse loop in the same car is not interrupted<br>**THEN**<br>"Gearbox overTemperature preAlarm detected by PT100" driver Warningis generated<br>Each event shall report the information about which car and axle is affected<br>The gear box PreAlarm Temperature Pictogram on the TOD shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.388 | **IF**<br>Gearbox Temp >= 175 °C for more than 30 s after a low pass filter with T=8sec:<br>**AND**<br>Thermal fuse loop in the same car is not interrupted<br>**THEN**<br>"Gear box overtemperature Alarm detected by PT100" C2 Serious Fault is generated<br>The information about which bogie and which gearbox is affected shall be reported on the TOD<br>The gear box Alarm Temperature Pictogram on the TOD shall be lit<br>The HABD desk lamp shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 2364 |

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| 2F_03.Zefiro-Europe.CONCEPT.364 | TCMS shall monitor single shaft of each **gearbox** of the trainset and set following events based on the difference between the two temperature sensors.<br>The delta temperature for each shaft will be calculated only if the temperatures of both bearing of the shaft are > 40°C.<br><br>**Motor side bearings (fast side)**<br>– Maintenance message: Difference between both sensors \|T1-T2\|>30°C for more than 30 s, after a low pass filter with T=8sec, maintenance message will be issued to the maintainer.<br>**Wheelset side bearing (slow side)**<br>– Maintenance message: Difference between both sensors \|T1-T2\|>25°C for more than 30 s,after a low pass filter with T=8sec, maintenance message will be issued to the maintainer.<br><br>**IF** the above maintenance condition are verified<br>**AND**<br>the car loop is not interruped<br>**THEN**<br>"Gear box overtemperature preAlarm detected by PT100" Driver warning shall be generated | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |

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| 2F_03.Zefiro-Europe.CONCEPT.HR30 | TCMS shall monitor single shaft of each **gearbox** in the trainset and set following events based on the difference between the two temperature sensors.<br>The delta temperature for each shaft will be calculated only if the temperatures of both bearing of the shaft are > 40°C.<br><br>**Motor side bearings (fast side)**<br>– Alarm: Difference between both sensors \|T1-T2\|>40°C for more than 30 s, after a low pass filter with T=8sec, the Alarm will be issued on driver's IDU only if the warmest bearing of the shaft has temperature >100°C<br><br>**Wheelset side bearing (slow side)**<br>- Alarm: Difference between both sensors \|T1-T2\|>35°C for more than 30 s, after a low pass filter with T=8sec, the Alarm will be issued on driver's IDU only if the warmest bearing of the shaft has temperature >100°C<br><br>**IF** the above Alarm condition are verified<br>**AND**<br>the car loop is not interruped<br>**THEN**<br>"Gear box overtemperature Alarm detected by PT100" C2 Serious Fault is generated<br>Each event shall report the information about which axle in the car is affected and will require to the driver to stop immediately<br>The gear box Alarm Temperature Pictogram on the TOD shall be lit<br>The HABD desk lamp shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.377 | TCMS shall monitor single bearing of each **gearbox** in the trainset and set following events based on the gradient evaluated over 30 sec.<br>**IF**<br>-Alarm: gradient > 12,5°C where the starting temperature is >80°C<br>**AND**<br>Thermal fuse loop in the same car is not interrupted<br>**THEN**<br>"Gear box overtemperature Alarm detected by PT100" C2 serious Fault is generated<br>Each event shall report the information about which axle in the car is affected<br>The Gear box Alarm temperature Pictogram on the TOD shall be lit<br>The HABD desk lamp shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |

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| 2F_03.Zefiro-Europe.CONCEPT.HR04 | **IF**<br>a **gearbox** overtemperature pre Alarm (Req 2F_03.Zefiro-Europe.CONCEPT.HR02,.HR03,…) or C2 Serious Fault condition (Req 2F_03.Zefiro-Europe.CONCEPT.388, ..377) is verified for a gear box in the trainset<br>**AND**<br>Thermal fuse loop in the same car is interrupted<br>**THEN**<br>"Gear box excessive temperature HABD Alarm detected by Thermal Fuse" C1 Serious Fault is generated<br>Each event shall report the information about which axle in the car is affected<br>The Gear box Alarm temperature Pictogram on the TOD shall be lit<br>The HABD Desk lamp shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.HR19 | **IF**<br>a **gearbox** temperature signal, evaluated after a low pass filter with a 8 sec time costant, exceeds +184°C for more than 30 s<br>**THEN**<br>"**gearbox** excessive temperature HABD Alarm detected by PT100" C1 Serious Fault is generated<br>Each event shall report the information about which axle in the car is affected<br>The axlebox Alarm temperature Pictogram on the TOD shall be lit<br>The HABD Desk lamp shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.117 | **IF**<br>the **axlebox** temperature signal, evaluated after a low pass filter with a 8 sec time costant, exceeds +95 °C for more than 30 s,<br>**AND**<br>Thermal fuse loop in the same car is not interrupted<br>**THEN**<br>"Axlebox overTemperature preAlarm detected by PT100" warning to driver is generated<br>Each event shall report the information about which car and bearing is affected<br>The Axlebox PreAlarm temperature Pictogram on the TOD shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.362 | **IF**<br>an **axlebox** temperature signal evaluated after a low pass filter with a 8 sec time costant, exceeds +110°C for more than 30 s<br>**AND**<br>Thermal fuse loop in the same car is not interrupted<br>**THEN**<br>"Axlebox overTemperature Alarm detected by PT100" C2 Serious Fault is generated<br>Each event shall report the information about which car and bearing is affected<br>The Axlebox Alarm temperature Pictogram on the TOD shall be lit.The HABD desk lamp shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 2 |

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| 2F_03.Zefiro-Europe.CONCEPT.HR05 | **IF**<br>One of the **axlebox** overtemperature pre Alarm (Req 2F_03.Zefiro-Europe.CONCEPT.117) or C2 Serious Fault temperature condition (Req 2F_03.Zefiro-Europe.CONCEPT.362) is verified<br>**AND**<br>Thermal fuse loop in the same car is interrupted<br>**THEN**<br>"Axle box excessive temperature HABD Alarm detected by Thermal Fuse" C1 Serious Fault is generated<br>Each event shall report the information about which bogie and which gearbox is affected<br>The Axle box Alarm temperature Pictogram on the TOD shall be lit<br>The HABD Desk lamp shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.HR17 | **IF**<br>an **axlebox** temperature signal, evaluated after a low pass filter with a 8 sec time costant, exceeds +121°C for more than 30 s<br>**THEN**<br>"Axle box excessive temperature HABD Alarm detected by PT100" C1 Serious Fault is generated<br>Each event shall report the information about which bogie and which gearbox is affected<br>The Axle box Alarm temperature Pictogram on the TOD shall be lit<br>The HABD Desk lamp shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.HR18 | **IF**<br>an **axlebox** temperature signal, evaluated after a low pass filter with a 8 sec time costant, exceeds +121°C for more than 30 s or **a gear box** temperature exceeds 184°C for more than 30 sec<br>**AND**<br>Thermal fuse loop in the same car is not interrupted<br>**THEN**<br>"thermal fuse loop incoherent with PT100" warning to driver is generated<br>the event shall report the information about which car is affected. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.359 | If<br>– RIO is on AND RIO declares a nth PT100 input as valid.<br>– independently from train speed<br>the temperature of all **gearboxes** shall be evaluated:<br>If the temperature is out of range ( <-50°C or >500°C ) the sensor shall be considered in failure The above condition last until PT100 temperature are back into the plausibility range. | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.HR31 | If a gearbox sensor temperature is in failure a warning diagnostic event shall be generated inadicating the identification of the affected gearbox PT100 | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.241 | If<br>– RIO is on AND RIO declares a nth PT100 input as valid.<br>– independently from train speed | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 2 |

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|---|---|---|---|---|---|
| | the temperature of all **axle boxes** shall be evaluated:<br>If the temperature is out of range ( <-50°C or >500°C ) the sensor shall be considered in failure The above condition last until PT100 temperature are back into the plausibility range. | | | | |
| 2F_03.Zefiro-Europe.CONCEPT.HR32 | If an axlebox sensor temperature is in failure a warning diagnostic event shall be generated inadicating the identification of the affected axlebox PT100 | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |

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|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.371 | **PT100** gradient assessment to declare **excluded the gearbox bearing Pt100** sensor temperature**.**<br><br>1) Ratio of gradient of PT100 bearing temperature when negative:<br>– IF a cab is activated<br>– IF the temperature gradient of nth bearing of mth gearbox is <0<br>– AND if the ratio between the module of the temperature gradient of the nth bearing and the biggest module among all the remaining bearings on the same bogie (having temperature gradient < 0) of the trainset is >=4,<br>– AND this condition shall be repeated for 3 times consecutively<br>the PT100 of the nth bearing has to be declared as ***excluded type C*** and their temperature readings will be discarded.<br>A PT100 declared as excluded type C will stay in this condition until next train reset.<br><br>2) Gradient of PT100 bearing temperature when positive:<br>– IF a cab is activated<br>– IF the temperature gradient of nth bearing of mth gearbox is >0<br>– AND IF the temperature gradient of the nth bearing of mth gearbox is > 10 °C/s,<br>the PT100 of the nth bearing has to be declared as ***excluded type A*** and their temperature readings will be discarded.<br>A PT100 declared as ***excluded type A*** will stay in this condition until next train reset.<br><br>3) Gradient of PT100 bearing temperature when negative:<br>– IF a cab is activated<br>– IF the temperature gradient of nth bearing of mth gearbox is <0<br>– AND the temperature gradient of the nth bearing of mth gearbox is < -10 C°/s,<br>the PT100 of the nth bearing has to be declared as ***excluded type A*** and their temperature readings will be discarded.<br>A PT100 declared as ***excluded type A*** will stay in this condition until next train reset.<br><br>The above cases shall generate a maintenance event to report the identification of the affected gearbox bearing PT100<br><br>Note: the temperature gradient for the bullet 1 shall be calculated with dt of 10s | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |

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|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.376 | **PT100** gradient assessment to declare **excluded the axlebox bearing Pt100** sensor temperature**.**<br><br>1) Ratio of gradient of PT100 bearing temperature when negative:<br>– IF a cab is activated<br>– IF the temperature gradient of nth bearing of mth axlebox is <0<br>– AND if the ratio between the module of the temperature gradient of the nth bearing and the biggest module among all the remaining homologous bearings (having temperature gradient < 0) of the trainset is >=4,<br>– AND this condition shall be repeated for 3 times consecutively<br>the PT100 of the nth bearing has to be declared as ***excluded type C*** and their temperature readings will be discarded.<br>A PT100 declared as ***excluded type C*** will stay in this condition until next train reset.<br><br>2) Gradient of PT100 bearing temperature when positive:<br>– IF a cab is activated<br>– IF the temperature gradient of nth bearing of mth axlebox is >0<br>– AND IF the temperature gradient of the nth bearing of mth axlebox is > 10 °C/s,<br>the PT100 of the nth bearing has to be declared as ***excluded type A*** and their temperature readings will be discarded.<br>A PT100 declared as ***excluded type A*** will stay in this condition until next train reset.<br><br>3) Gradient of PT100 bearing temperature when negative:<br>– IF a cab is activated<br>– IF the temperature gradient of nth bearing of mth axlebox is <0<br>– AND the temperature gradient of the nth bearing of mth axlebox is < -10 C°/s,<br>the PT100 of the nth bearing has to be declared as ***excluded type A*** and their temperature readings will be discarded.<br>A PT100 declared as ***excluded type A*** will stay in this condition until next train reset.<br><br>The above cases shall generate a maintenance event to report the identification of the affected axlebox bearing PT100<br><br>Note: the temperature gradient for the bullet 1 shall be calculated with dt of 10s | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |

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|---|---|---|---|---|---|
| 2F_03. Zefiro-Europe.CONCEPT.HR08 (ex 2F_05.02.Zefiro-Europe.CONCEPT.4120) | **GIVEN** ( PT100_i data of the **axle box** is SW valid )<br><br>**WHEN**<br><br>(<br>$(PT100_i - \frac{1}{7}\sum_{\substack{j=1 \\ j \neq i}}^{8} PT100_j) \geq \Delta a$<br>)<br><br>**THEN** ( PT100_i value of the axle box is "***Plausible***" )<br><br>$\forall\, i = 1, \ldots, 8$<br>$\Delta a = -30\,°C$<br>The Pt100 signals in the previous relation are filtered with a low pass filter with a time constant of 8 sec | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |
| 2F_03. Zefiro-Europe.CONCEPT.HR33 | When an axlebox PT100 is evaluated not Palusible an event shall be generated | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03. Zefiro-Europe.CONCEPT.HR09 (ex 2F_05.02.Zefiro-Europe.CONCEPT.4129) | GIVEN ( PT100_j MIO data of the **big wheel shaft bearings** is SW valid )<br><br>**WHEN**<br><br>(<br>$(PT100_j - \frac{1}{7}\sum_{\substack{k=9 \\ j \neq k}}^{16} PT100_k) \geq \Delta g$<br>)<br><br>**THEN** ( PT100_j value of the big wheel shaft bearings is "***Plausible***" )<br><br>$\forall\, j = 9, \ldots, 16$<br>$\Delta g = -30\,°C$<br>The Pt100 signals in the previous relation are filtered with a low pass filter with a time constant of 8 sec | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |
| 2F_03. Zefiro-Europe.CONCEPT.HR10 (ex 2F_05.02.Zefiro-Europe.CONCEPT.4169) | GIVEN ( PT100_j data of the **pinion shaft bearings** is SW valid )<br><br>**WHEN**<br><br>(<br>$(PT100_j - \frac{1}{7}\sum_{\substack{k=17 \\ j \neq k}}^{24} PT100_k) \geq \Delta g$<br>)<br><br>**THEN** ( PT100_j value of the **pinion shaft bearings** is "***Plausible***" )<br><br>$\forall\, j = 17, \ldots, 24$<br>$\Delta g = -30\,°C$<br>The Pt100 signals in the previous relation are filtered with a low pass filter with a time constant of 8 sec | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL2 |

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|---|---|---|---|---|---|
| 2F_03. Zefiro-Europe.CONCEPT.HR34 | When a gearbox PT100 is evaluated not Palusible an event shall be generated | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03. Zefiro-Europe.CONCEPT.HR11 (ex 2F_05.02.Zefiro-Europe.CONCEPT.4142) | **GIVEN** ( PT100_i data of the **axle box** is SW valid )<br><br>**WHEN**<br><br>(<br>$abs((PT100_i - \frac{1}{7}\sum_{\substack{j=1 \\ j \neq i}}^{8} PT100_j)) < \Delta a$<br>)<br><br>**THEN** ( PT100_i value of the **axle box** is "***Plausible***" )<br><br>$\forall\, i = 1, \dots, 8$<br>$\Delta a = 30\ °C$<br>The Pt100 signals in the previous relation are filtered with a low pass filter with a time constant of 8 sec | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |
| 2F_03. Zefiro-Europe.CONCEPT.HR12 (ex 2F_05.02.Zefiro-Europe.CONCEPT.4144) | **GIVEN** ( PT100_j data of the big **wheel shaft bearings** is SW valid )<br><br>**WHEN**<br><br>(<br>$abs((PT100_j - \frac{1}{7}\sum_{\substack{k=9 \\ j \neq k}}^{16} PT100_k)) < \Delta g$<br>)<br><br>**THEN** ( PT100_j value of the **big wheel shaft bearings** is "***Plausible***" )<br><br>$\forall\, j = 9, \dots, 16$<br>$\Delta g = 30\ °C$<br>The Pt100 signals in the previous relation are filtered with a low pass filter with a time constant of 8 sec | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |
| 2F_03. Zefiro-Europe.CONCEPT.HR13 (ex 2F_05.02.Zefiro-Europe.CONCEPT.4168) | **GIVEN** ( PT100_j RIO data of the **pinion shaft bearing**s is SW valid )<br><br>**WHEN**<br><br>(<br>$abs((PT100_j - \frac{1}{7}\sum_{\substack{k=17 \\ j \neq k}}^{24} PT100_k)) < \Delta g$<br>)<br><br>THEN ( PT100_j value of the **pinion shaft bearings** is "***Plausible***" )<br><br>$\forall\, j = 17, \dots, 24$<br>$\Delta g = 30\ °C$<br>The Pt100 signals in the previous relation are filtered with a low pass filter with a time constant of 8 sec | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |

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|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.HR15 | **IF**<br>(<br>a gearbox PT100 sensor in the car is failure (2F_03.Zefiro-Europe.CONCEPT.359)<br>**OR**<br>a gearbox PT100 sensor in the car is excluded type A or Type C (2F_03.Zefiro-Europe.CONCEPT.371)<br>**OR**<br>a gearbox PT100 sensor in the car is NOT Plausible (req 2F_03.Zefiro-Europe.CONCEPT.HR09 and 2F_03.Zefiro-Europe.CONCEPT.HR10 and 2F_03.Zefiro-Europe.CONCEPT.HR12 and 2F_03.Zefiro-Europe.CONCEPT.HR13 not verified )<br>**)**<br>**OR**<br>2F_03.Zefiro-Europe.CONCEPT.364 is verified<br>**OR**<br>2F_03.Zefiro-Europe.CONCEPT.HR30 is verified<br>**AND**<br>Thermal fuse loop in the same car is interrupted<br>**THEN**<br>“Gearbox excessive temperature HABD Alarm detected by Thermal Fuse” C1 Serious Fault is generated<br>Each event shall report the information about which axle in the car is affected<br>The gearbox Alarm temperature Pictogram on the TOD shall be lit<br>The HABD Desk lamp shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.HR14 | **IF**<br>(<br>an axle box PT100 sensor in the car is failure (2F_03.Zefiro-Europe.CONCEPT.241)<br>**OR**<br>a axle box PT100 sensor in the car is excluded type A or Type C (2F_03.Zefiro-Europe.CONCEPT.376)<br>**OR**<br>a axle box PT100 sensor measure in the car is NOT Plausible (req 2F_03.Zefiro-Europe.CONCEPT.HR08 and 2F_03.Zefiro-Europe.CONCEPT.HR11 not verified )<br>**)**<br>**AND**<br>Thermal fuse loop in the same car is interrupted<br>**THEN**<br>“Axle box excessive temperature HABD Alarm detected by Thermal Fuse” C1 Serious Fault is generated<br>Each event shall report the information about which bearing in the car is affected<br>The Axle box Alarm temperature Pictogram on the TOD shall be lit<br>The HABD Desk lamp shall be lit | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |

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|---|---|---|---|---|---|
| Europe.CONCEPT.HR16 | **IF**<br>TCMS detects the de-energization of the HABD car loop in the trainset<br>**AND**<br>No axle box or gearbox Pt100 in the car indicates a preAlarm or Alarm condition<br>**AND**<br>No axle box or gearbox Pt100 in the car is failure or excluded or indicates a not plausible value<br>**THEN**<br>"Undue HABD car loop interruption"Warning to driver is generated | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.389 | Every time any Driver warning (both for axleboxes and garboxes) is set the related pictogram shall be shown on IDU. If a Serious Fault C2 or C1 follows a warning activation of the same criteria on the same sensor the transition between the pre-alarm and Serious Fault pictogram has to be continuous. | Derived Requirement | 4S_09.03.05.-.TCMS Software | IDU | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.380 | Any axle box or gear box temperature Serious Fault shall be latched (and so memorized also if the setting conditions are cleared).<br><br>The memorization will be cancelled only by pressing a **dedicated Safe softkey** "HABD Reset" on the related bogie page (available only at train standstill) and the alarm conditions are not active. | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.381 | The use of the HABD reset softkey shall be recorded as an event. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.378 | If any gearbox or axle box overtemperature C2 Serious Fault is triggered the TCMS shall activate a permanent traction block as long as the train speed >= 80 km/h.<br><br>If the train speed is < 80 km/h the traction block can be restored (after positioning the traction lever in coasting). | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.383 | In case the driver has reset the C2 Serious Fault (not valid for C1 Serious Fault) the speed limitation shall not be active anymore even if the alarm setting conditions are still valid. | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.384 | In case of activation of a new C2 Serious Fault (from another PT100 sensor) OR in case of re-activation of the the cleared alarm the speed limit shall be again active. | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.386 | Any pre-alarm and/or alarm shall be shown on the wayside interface together with its "acknowledged/not acknowledged" information (additional checkbox). | Derived Requirement | 4S_09.03.01.01.-.Train To Wayside Communication Systems | | SIL 0 |
| | | | | | |
| 2F_03.Zefiro-Europe.CONCEPT.HR22 | During HABD test test the buzzer shall be not sound. | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.HR23 (ex 2F_05.02.Zefiro-Europe.CONCEPT.4138) | **GIVEN**<br><br>"HABD Alarm" C1 or C2 Serious Fault is active | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 2 |

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|---|---|---|---|---|---|
| | **WHEN**<br>(<br>( the dedicated *Safe Key* "HABD Reset" on IDU is pushed )<br>**AND**<br>( Vehicle at "*Standstill*" and the alarm cause is not present)<br>)<br>**THEN** ( " All *HABD* Serious Fault are reset) | | | | |
| 2F_03.Zefiro-Europe.CONCEPT.HR28 | IF<br>(<br>All the *HABD Alarm*" C1 or C2 Serious Fault are reset<br>)<br>THEN<br>(<br>The HABD desk lamp shall be switched off<br>) | | 4S_09.03.05.-.TCMS Software | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.HR29 | IF<br>(<br>All the *HABD Alarm*" C1 or C2 Serious Fault are reset<br>)<br>THEN<br>(<br>The HABD pictograms shall be switched off<br>) | | 4S_09.03.05.-.TCMS Software | | SIL 0 |

*Test Function*

| | |
|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.108 |
| **Description** | A test function shall be implemented for the hot box detection |
| **Safety level** | |

To ensure the functionality of the hot axle box detection safety loop a test function shall be integrated.

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.339 | A relay shall be installed in each car and shall trigger a "break" contact of each bogie of this car. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | |
| 2F_03.Zefiro-Europe.CONCEPT.343 | The test relay shall be linked to TCMS digital output. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | |
| 2F_03.Zefiro-Europe.CONCEPT.344 | On TOD the HABD test button shall be present. If the button is pressed TCMS shall start the HABD test sequence on the whole trainset. | Derived Requirement | 4S_09.03.05.-.TCMS Software | | |
| 2F_03.Zefiro-Europe.CONCEPT.315 | A permanent supervision of the HABD car loop relays shall be implemented in TCMS<br>If one supervision signal shows a different state of the relay than the other for more than 1 second, an EVENT shall be set in the diagnostic system stating implausibility of the hot box detection safety relays. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |
| 2F_05.02.Zefiro-Europe.CONCEPT.HR25 | The HABD test sequence consist of to energize the BMS test relays car by car in sequence.<br>In multiple composition , the test sequence shall be executed in parallel on both train | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |

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| | During this operation TCMS shall check following status:<br>– The HABD safety loop is opened (reading HABD relays status) car by car | | | | |
| 2F_05.02.Zefiro-Europe.CONCEPT.HR26 | The HABD test shall be passed if the safety loop results opened by reading both the HABD relay contacts status.and then reclosed when the test signal has been reset | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_05.02.Zefiro-Europe.CONCEPT.HR27 | If the BMS test has been passed succesful an event shall be set. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.347 | The result of the HABD-test shall be presented to the driver on the IDU. A failed test indication on the IDU shall be reset by a successful additional test. | Derived Requirement | 4S_09.03.05.-.TCMS Software | IDU | |
| 2F_03.Zefiro-Europe.CONCEPT.370 | The status of HABD loop relays shall be presented, per car, to the driver on the IDU. | Derived Requirement | 4S_09.03.05.-.TCMS Software | IDU | |

# Instability Monitoring

## Function Design

The Instability monitoring system supervises the lateral acceleration of a bogie during the service of the V300ZEFIRO – ETR1000 high speed train.
The bogie state that shall be considered as instability is defined in the HS RST TSI and implemented in the presented system. The detection is based on the measurement of the lateral acceleration of the bogie frame above the centreline of the wheelset.
In case the defined conditions are present (10 consecutive lateral acceleration cycles of 0,8g and a frequency between 3 and 9 Hz) the bogie mounted supervision devices (IMS boxes) indicate instability.
For availability reasons two IMS boxes are mounted diagonally on each bogie. They are stand-alone systems which provide via digital output to TCMS and via safety relays that open one of the train lines the detected instability status.
According to HS RST TSI the driver is informed by a visual signal on the drivers desk in case of instability and in addition to that a buzzer is sounded inside the cab. The activation of the indication to the driver is monitored by TCMS

The function of the IMS is to indicate instability to the driver when the TSI criteria for instability is exceeded. As soon as instability is present the driver is informed via the IDU.
For diagnostic purposes the type of instability is divided into 3 levels.

- Level 0
- Level 1
- Level 2

The Level 0 warning indicates short term instability cases which are present not longer than 1 second. This cases in which the lateral acceleration amplitude drops below the TSI relevant value of 0,8g within 1 second lead to investigative maintenance actions for bogie parts and the detection equipment.
Level 1 and Level 2 identify longterm (more the 1 second) instability cases. If the magnitude of the instability is below 15 m/s² the incident is stored in TCMS as level 1 Instability. If the magnitude is higher than 15 m/s², the incident is stored as level 2.

### *Normal condition*

Instability is detected by both IMS-units of a bogie and a instability warning or alarm is displayed on the driver's desk according to the specification of the warning levels.

### *Failure cases*

1. Instability is present but not detected.
2. No instability is present, but falsely indicated by one box of the bogie?

### *Failure detection*

1. Not possible to detect by the driver. In case of instability detection by the train crew(noise, vibration) train driver must be alerted.
2. It must be considered as an 'Instability detection', eventual IMS fault can be detected during IMS test

### *Failure handling*

1. Effected bogie has to be checked during corrective maintenance
2. Train acts according to the level. Boxes have to be repaired during unplanned workshop visit.

### *Failure effect on operation*

1. No effect from train. Driver has to act according to operational manual(actions have to be aligned with customer)
2. Train is limited to 220 km/h

## Requirements

This part describes the functions of the instability monitoring system unit that is located in the bogie and are related to all instability level warnings described below.

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|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.82 |
| **Description** | An on board instability monitoring system shall be fitted. The detection of instability shall be based on acceleration measurements taken on the bogie frame. An instabilty warning or alarm shall advise the driver to reduce the speed. The criteria for activation of this alarm shall be as defined in <<HS RST TSI 2008>> clause 4.2.3.4.2 d2. |
| **Safety level** | SIL 2 |

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.65 | An on board instability monitoring system with 2 redundant detection units per bogie shall be | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 2 |

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|---|---|---|---|---|---|
| | installed.The units shall be installed diagonally in the bogie. | | | | |
| 2F_03.Zefiro-Europe.CONCEPT.169 | An IMS unit shall include a lateral accelerometer that shall be placed on the bogie frame above the centreline of the wheelset. | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.170 | A lateral acceleration of a>=8 m/s² at a frequency between 3 to 9 Hz and for more than 10 consecutive cycles shall be considered as instability by the IMS unit and 'Instability Detection' shall be set. | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 2 |

| | |
|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.95 |
| **Description** | Two digital inputs shall be provided to indicate 4 states to TCMS:<br>– System OK<br>– High Level Detection (level 2)<br>– Instability Detection (level 0 and level 1)<br>– System Error |
| **Safety level** | SIL 0 |

Apart from the link to the trainline a IMS unit shall be connected to TCMS via 2 digital signals which provide 4 states to TCMS.

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.129 | 2 Digital inputs per IMS unit shall provide the IMS status to TCMS:<br>– System OK (1 1)<br>– High Level Instability Detection (level 2) (1 0)<br>– Instability Detection (level 0 and level 1) (0 1)<br>– System Error (0 0) | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.234 | The three levels of IMS indication shall be sent to TCMS via the I/O link:<br>– System OK<br>– High Level Instability Detection (level 2)<br>– Instability Detection (level 0 and level 1)<br>– System error | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.179 | System Error shall be set when an internal error of any kind occured in the IMS unit | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.181 | High Level Detection shall be set when instability Detection is active and a lateral acceleration peak of a>=15 m/s² occured. | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.182 | System OK shall be set if non of the following<br>– System Error or<br>– High Level Detection or<br>– Instability Detection<br>are set. | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.177 | 2 digital inputs and 1 output shall be linked from every IMS unit (per bogie) to TCMS | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 0 |

| | |
|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.91 |
| **Description** | For IMS, the detection threshold levels and number of consecutive cycles shall be adaptable. |
| **Safety level** | N/A |

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.120 | The detection treshold levels (lateral acceleration high detection level and low detection level) and the number of consecutive cycles of the IMS system shall be adaptable. | Derived Requirement | 4S_03.-.Bogies and Running Gear | | |

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|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.93 |
| **Description** | For availability reasons there shall be two redundant train-lines and two IMS units per bogie. The two IMS units per bogie shall be connected to different train lines. |
| **Safety level** | N/A |

| ID | 2F_03.Zefiro-Europe.TRS.92 |
|---|---|
| Description | The safety relays of all IMS units of the train shall be connected in serial connection to respective train-lines to communicate the instability warning to a lamp (and/or buzzer) in the cab. The trainline shall be of fail safe type i.e. the trainline is de-energised by the safety relays breaking the loop. |
| Safety level | SIL 2 |

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.123 | 2 trainlines shall be integrated in train. They shall connect the IMS boxes according to the schematic in 2.1.3. to the indication lamps on the drivers desk | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 2 |

The two trainlines shall be called "loop 1" and "loop 2" according to the schematic in 2.1.3.

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.233 | Both IMS loops shall be supervised by TCMS. If a loop is open, a supervision signal shall be set. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.215 | If the "loop 2" supervision signal is set and no IMS box shows instability, an event shall be set in the diagnostic system. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.331 | If the "loop 1" supervision signal is set and no IMS box shows instability, an event shall be set in the diagnostic system. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.321 | DELETED | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.322 | If both loops are open<br>or<br>If any IMS box shows "high level detection" or "instabiliy detection"<br>or<br>If both IMS boxes on a single bogie shows "System error"<br><br>it shall be considered as instability by TCMS and the 'Instability present' signal must be set | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.373 | In case the status of IMS controller indicates High level instability Detection (1 0) OR Instability detection (0 1) AND the related IMS loop is energized for more than 30 s,the IMS box shall be excluded by TCMS (considered in "System Error") and the status shall be sent to IDU<br>NOTE: the affected IMS box will stay in this condition until next train reset | Derived Requirement | 4S_09.03.05.-.TCMS Software | | |
| 2F_03.Zefiro-Europe.CONCEPT.231 | Two indication lamps shall be fitted in the driver's desk according to the schematic 2.1.3. | Derived Requirement | 4S_08.07.01.-.Drivers Desk | HW HMI | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.175 | The trainline shall be energised when no instability is detected. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.176 | If both trainlines are de-energised the lamp in the activated cab shall be active. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.174 | If instability is detected by an IMS box, the trainline must be broken by a safety relay in the IMS box when the additional requirements of level 1 or level 2 instability are fullfilled. | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.353 | If any IMS box indicates 'high level indication' or 'instability indication' a diagnostic event shall be set (indicating the affected box and its status) | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.374 | If any IMS box indicates "system error" | Derived Requirement | 4S_09.03.05.-.TCMS Software | IDU | |

| | | | | | |
|---|---|---|---|---|---|
| | an warning event shall be shown to the Driver (indicating the affected box and its status) | | | | |

## Level 0 Warning

### *Function Design*

As irregularities of the track may lead to short lateral oscillations of the bogie a level 0 warning was introduced into the IMS detection system. This state is only valid for 1 second.
When instability was detected by any IMS box, it is indicated to the train via the digital outputs of the boxes. The signal is read by TCMS and activates the instability alarm on the IDU.
After 1 second the safety relays in the IMS boxes open the train lines and the indication is escalated to a "Level 1" for which the actions are defined in section 3.3.4.
The following actions are performed during Level 0 warning:

- If instability is detected by an IMS box, the "Low level Detection" or "High level Detection" signal is set.
- If the status is still present after 1s, the safety relays of the affected IMS box are opened and the connected IMS train-line is deenergised.
- If any IMS box indicates "Low level Detection" or "High Level Detection" TCMS activates the instability alarm on the IDU and stores the Level 0 warning in the diagnostic system of the train including time and identification of the affected bogie.

### *Requirements*

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|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.68 |
| **Description** | Instability warning level 0 shall be reported via diagnostics (TCMS) if a lateral acceleration of low level is detected (Fe => 8m/s² is detected between 3Hz and 9Hz for 10 consecutive cycles). The driver shall have the possibilty to decide upon actions considering the track status. |
| **Safety level** | SIL 0 |

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.212 | If instability is detected a timer shall be started in the IMS unit | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.178 | If<br>– if the instability present signal is set (as described in 2F_03:Zefiro-Europe.CONCEPT.322)<br><br>and<br><br>– and the trainspeed is >220 km/h<br><br>a instability alarm shall be displayed on the IDU for the driver including the identification of the affected car and the buzzer of the activated cab shall be sound.<br>The event shall be stored in the diagnostic system of the train including time and identification.<br>If trains speed is above 220km/h the traction shall be blocked as long as the instability is present (as described in 2F_03:Zefiro-Europe.CONCEPT.178). | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.183 | If the instability present signal is set and train speed is below 220km/h an event shall be set in the diagnostic system | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |

## Level 1 Alarm

### *Function Design*

To improve the reliability of the system two train-lines are implemented in the train. Each train-line connects in serie 16 IMS boxes (one out of two per bogie) to a relay in the cab. If instability is present on a bogie and is detected for more than 1 second on both IMS boxes of the bogie (see Level 0 warning in section 3.3.3), both train-lines are opened and lead to the activation of the indication lamp on the drivers desk and the activation of the main buzzer inside the cab.
This logic implements that the faulty indication of instability by one out of two IMS boxes per bogie does not lead to the activation of the indication devices on the drivers desk.

Only in case of a double failure (failure of one IMS box on both train-lines even if not on the same bogie) instability is falsely indicated to the driver when no instability is present.
The following actions are performed during Level 1 Alarm:

- The detection of instability for more than 1 second on one IMS box per train-line leads to the opening of both IMS train-lines
- The opening of both train-lines lead to the strictly wired activation of the indication lamp (see 2.1.3.).
- If both train-lines are opened TCMS sets the 'instability present' signal on the base of the digital inputs for **loop1** and **loop2** (see Figure 2.1.3.)
- When the "instability present" signal is set all bogies having at lease one "Low Level Detection" active are considered as affected by Level 1 instability.
- The identification of the affected bogie(s) is displayed on the IDU.
- The status "Level 1", the identification of the affected bogie(s) and the time is stored in the diagnostic system.
- The activation of the "instability present" signal is stored within TCMS.

*Requirements*

| ID | 2F_03.Zefiro-Europe.TRS.69 |
|---|---|
| Description | Instability warning level 1 shall be indicated if the timer for an active level 0 warning exceeds 1s. This warning - now escalating into an alarm - shall be reported via the IMS trainlines. |
| Safety level | SIL 2 |

| ID | 2F_03.Zefiro-Europe.TRS.103 |
|---|---|
| Description | Instability warning level 1 shall be indicated if the timer for an active level 0 warning exceeds 1s. This warning - now escalating into an alarm - shall be reported via the diagnostic system (TCMS). |
| Safety level | N/A |

A safety relay shall be integrated in the IMS unit.

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.213 | If<br>– instability is detected and<br>– the IMS unit timer exceeds 1s<br>the IMS unit safty relay shall break the IMS trainline. | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 2 |

If

– if the instability present signal is set (as described in 2F_03:Zefiro-Europe.CONCEPT.322)

and

– both IMS trainlines are open

and

– and the train speed is >220 km/h

an instability alarm shall be displayed on the IDU for the driver including the identification of the affected car and the buzzer of the activated cab shall be sound.
The event shall be stored in the diagnostic system of the train including time and identification.
If trains speed is above 220km/h the traction shall be blocked as long as the instability is present(as described in 2F_03:Zefiro-Europe.CONCEPT.178).

## Level 2 Alarm

*Function Design*

Indication of instability "Level 2 Alarm"
If the lateral acceleration detected by an IMS box rises to 15m/s² the "High Level Detection" is set by the IMS box. The functionality for the activation of the indication devices is not affected by the Level of the Alarm.
The following actions are performed during Level 2 Alarm:

- When the “instability present” signal is set all bogies having at lease one “High Level Detection” active are considered as affected by Level 2 instability.
- The identification of the affected bogie(s) is displayed on the IDU.
- The status “Level 2 Alarm”, the identification of the affected bogie(s) and the time is stored in the diagnostic system.

*Requirements*

| ID | 2F_03.Zefiro-Europe.TRS.77 |
|---|---|
| Description | If in the mean time a peak acceleration is detected by IMS in excess of 15 m/s² (high level detected), then the alarm shall be reported via:<br>– - the IMS trainline being deenergised |
| Safety level | SIL 2 |

| ID | 2F_03.Zefiro-Europe.TRS.104 |
|---|---|
| Description | If in the mean time a peak acceleration is detected by IMS in excess of 15 m/s² (high level detected), then the alarm shall be reported via the IMS delivering the signal "high level" (level 2) to TCMS |
| Safety level | SIL 0 |

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.80 | if<br>– instability is detected and<br>– in the mean time a lateral acceleration of a>15m/s² is detected once<br>the IMS unit safty relay shall break the trainline | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.275 | if<br>– instability is detected and<br>– in the mean time a lateral acceleration of a>15m/s² is detected once<br>the High Level Detection shall be set | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 0 |

If

– if the instability detection signal is set (as described in 2F_03:Zefiro-Europe.CONCEPT.322)

and

– both IMS trainlines are open

and

– one or both IMS boxes of a single bogie show "high level detection"

and

– the train speed is >220 km/h

a High level instability alarm event shall be set including the identification of the affected car and the buzzer of the activated cab shall be sound.
The event shall be stored in the diagnostic system of the train including time and identification.
If trains speed is above 220km/h the traction shall be blocked as long as the instability is present (as described in 2F_03:Zefiro-Europe.CONCEPT.178)

| ID | 2F_03.Zefiro-Europe.TRS.76 |
|---|---|
| Description | If the instability level 2 alarm is triggered, train speed shall be limited. If speed is higher than the speed limit when the alarm is triggered, |
| Safety level | SIL 0 |

# Test

## *Function Design*

### *Requirements*

| ID | 2F_03.Zefiro-Europe.TRS.44 |
|---|---|
| Description | The IMS shall perform an automatic self-test after Power-up |
| Safety level | SIL 2 |

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.95 | After power-up a selftest shall be started by every IMS unit. Including test of<br>– sensors<br>– detection algoritm<br>– safety relays | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.221 | when a failure is detected by the self test - system error - shall be set by the effected unit and the safety relay shall break the trainline. | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 2 |
| 2F_03.Zefiro-Europe.CONCEPT.223 | if system error is set for TCMS must log the fault. This requirement is independent from the way the test is triggered. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | SIL 0 |

| ID | 2F_03.Zefiro-Europe.TRS.48 |
|---|---|
| Description | A IMS self test shall be triggered by TCMS if a cab is activated and the train is standstill. |
| Safety level | SIL 0 |

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.141 | When an end cab is activated and the train is at standstill, the selftest of the IMS units shall be activated by setting the test DO for both IMS boxes on a bogie at same time and checking the status of the related IMS loop. Each Car of the trainset shall be checked one by one.<br>TCMS shall trigger the test if at least one BC2 contactor is detected closed in each car. | Derived Requirement | 4S_09.03.05.-.TCMS Software | | SIL 0 |
| 2F_03.Zefiro-Europe.CONCEPT.235 | If the signal to trigger a IMS self test is received from TCMS, the IMS shall perform its self test. | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 0 |

| ID | 2F_03.Zefiro-Europe.TRS.52 |
|---|---|
| Description | The IMS train line during shall be open during test of the system. |
| Safety level | SIL 2 |

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.144 | During a IMS selftest the trainline shall be opened. | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 2 |

| ID | 2F_03.Zefiro-Europe.TRS.94 |
|---|---|
| Description | Every part shall be tested simultaneously on all IMS systems of the train. If the test is not successful, the driver is informed via the IMS lamp associated with the loop. |
| Safety level | SIL 2 |

| ID | 2F_03.Zefiro-Europe.TRS.101 |
|---|---|
| Description | If the IMS test is not successful, the driver shall be informed via the diagnostic system (TCMS). |
| Safety level | N/A |

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.147 | When a test is triggered, every part shall be tested simultaneously on all IMS systems of the train. If the test is not succesful, the driver shall | Derived Requirement | 4S_03.-.Bogies and Running Gear | | SIL 2 |

| | | | | | |
|---|---|---|---|---|---|
| | be informed via the IMS lamp associated with the loop. | | | | |
| 2F_03.Zefiro-Europe.CONCEPT.237 | The status of the IMS test shall be reported to TCMS using digital inputs of the TCMS | Derived Requirement | 4S_03.-.Bogies and Running Gear | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.236 | If the IMS test has not been successful (One IMS box or more states system error, IMS loop 1 does not open, IMS loop 2 does not open), an event shall be triggered in the diagnostic system. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |

# Active Lateral Suspension (ALS)

## Function Design

The objective of the ALS function is to improve the comfort level of the train by active supervising the carbody position compared to the bogie and stabilize a centre position. It is managed by the bogie controller, a stand alone system which is connected to the TCMS to transmit diagnostic, configuration and status info and to receive the speed level, 400V present and the train configuration. The bogie controller manages the ALS system on both bogies of the same car. Additional to the active lateral suspension, each bogie is equipped with lateral buffers to assure running stability and safety in case of a failure of the ALS. That means, that ALS is not safety relevant.

### *Normal condition*

The ALS has basically two tasks:

1. To create a lateral force in curves corresponding to lateral quasi-static acceleration times the pivot load of the bogie.
2. To reduce the vibration level in the carbody by creating a force counteracting the vibration transfer from bogie to carbody

### *Failure cases*

1. Electrical power of the actuator lost - the actuator operates as a passive lateral damper
2. Zero force reference - the actuator will minimize force
3. Maximum force to one side - actuator will force bumbstop contact

### *Failure detection*

1. tbd. - condition not safety relevant according to [6]
2. tbd. - condition not safety relevant according to [6]
3. tbd. - condition not safety relevant according to [6]

### *Failure handling*

1. Repair during next scheduled maintenance
2. Repair during next scheduled maintenance
3. Repair during next scheduled maintenance

## Requirements

| | |
|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.88 |
| **Description** | An ALS shall be provided on all bogies to ensure a good running behaviour and comfort. |
| **Safety level** | SIL 3 |

| | |
|---|---|
| **ID** | 2F_03.Zefiro-Europe.TRS.96 |
| **Description** | The ALS is integrated in the bogie controller which shall be connected to TCMS Ethernet to receive train data (e.g. speed) and to provide diagnostics and status reporting. |
| **Safety level** | SIL 3 |

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.195 | The following status and configuration signals are provided to the bogie contoller (relevant for the ALS):<br>– train speed<br>– Drive direction<br>– logical car number<br>– car orientation<br>– first car of the consist<br>– 400 V AC is present<br>– tbd in ICD document | Derived Requirement | 4S_09.03.05.-.TCMS Software | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.196 | The following events related to ALS shall be triggerd depending on inputs from the bogie controller | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |

| | | | | | |
|---|---|---|---|---|---|
| | – status data<br>– diagnostic data<br>– configurationinfo<br>of all ALS components (to be stored by TCMS, events tbd) | | | | |
| 2F_03.Zefiro-Europe.CONCEPT.199 | IP link from Bogie controller to TCMS | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | |

# Diagnostics & CBM

## Function Design

The CBM Controller monitors the conditions of certain bogie components to optimise the maintenance interventions by detecting degradations and failures.

## Requirements

| ID | 2F_03.Zefiro-Europe.TRS.79 |
|---|---|
| **Description** | The conditions of the bogie components shall be monitored continuously by a CBM system to optimise the behaviour in maintenance by early detecting degradations of e.g axle bearings, wheels and axles. |
| **Safety level** | N/A |

| ID | 2F_03.Zefiro-Europe.TRS.60 |
|---|---|
| **Description** | Periodical measurement analysis shall be done for the following components: bearings, wheels and axles. Notifications shall be given to the maintenance regime and if necessary also to the driver. |
| **Safety level** | SIL 0 |

CBM data is provided by the bogie monitoring equipment on the Ethernet to TCMS or directly to the Telediagnostica application.

| Nr | Description | Applicability Details | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.104 | The following data shall be provided to the Bogie Monitoring:<br>– Trainspeed<br>– train configuration, number of cars<br>– GPS position | Derived Requirement | 4S_09.03.05.-.TCMS Software | | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.206 | The following events shall be generated by TCMS:<br>– tbd | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_03.Zefiro-Europe.CONCEPT.238 | The following data from the bogie controller shall be handled:<br>– real time process data to Telediganostica application<br>– CBM data<br>– tbd | Derived Requirement | 4S_09.03.05.-.TCMS Software | | N/A |

# 4Signal Interface

## I/O Interface

### Supplier system

Inputs:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.290 | WFL: Enable wheel flange lubrication | L | 2F_03.Zefiro-Europe.CONCEPT.157<br>2F_03.Zefiro-Europe.CONCEPT.350 |

Outputs:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.29 | WFL: Spray impuls | L | 2F_03.Zefiro-Europe.CONCEPT.167 |
| 2F_03.Zefiro-Europe.CONCEPT.291 | WFL: System fault | L | 2F_03.Zefiro-Europe.CONCEPT.159<br>2F_03.Zefiro-Europe.CONCEPT.351 |
| 2F_03.Zefiro-Europe.CONCEPT.297 | IMS: 2 Digital outputs | L | 2F_03.Zefiro-Europe.CONCEPT.129 |

### TCMS SW

Inputs:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.30 | WFL: Spray impuls | L | 2F_03.Zefiro-Europe.CONCEPT.280 |
| 2F_03.Zefiro-Europe.CONCEPT.352 | WFL: System fault | L | 2F_03.Zefiro-Europe.CONCEPT.351 |
| 2F_03.Zefiro-Europe.CONCEPT.294 | HABD: Bogie supervision loop energized | L | 2F_03.Zefiro-Europe.CONCEPT.257 |

HABD: Supervision of safety relais 1
HABD: Supervision of safety relais 2

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.295 | HABD: Temperatur + ID of axle box | A | 2F_03.Zefiro-Europe.CONCEPT.257 |
| 2F_03.Zefiro-Europe.CONCEPT.296 | IMS: 2 Digital inputs | L | 2F_03.Zefiro-Europe.CONCEPT.129<br>2F_03.Zefiro-Europe.CONCEPT.214 |
| 2F_03.Zefiro-Europe.CONCEPT.298 | IMS: Supervision of IMS loop 2 | L | 2F_03.Zefiro-Europe.CONCEPT.215 |
| 2F_03.Zefiro-Europe.CONCEPT.337 | IMS: Supervision of IMS loop 1 | L | 2F_03.Zefiro-Europe.CONCEPT.331 |
| 2F_03.Zefiro-Europe.CONCEPT.332 | IMS: Instability present | L | 2F_03.Zefiro-Europe.CONCEPT.214<br>2F_03.Zefiro-Europe.CONCEPT.322 |
| 2F_03.Zefiro-Europe.CONCEPT.333 | IMS: Instability loop at fault | L | 2F_03.Zefiro-Europe.CONCEPT.321 |

Outputs:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.293 | WFL: enable wheel flange lubrication system | L | 2F_03.Zefiro-Europe.CONCEPT.108<br>2F_03.Zefiro-Europe.CONCEPT.350 |
| 2F_03.Zefiro-Europe.CONCEPT.292 | WFL: activate solinoid valve | L | 2F_03.Zefiro-Europe.CONCEPT.280 |

## Bus Interface

Supplier to TCMS SW:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.32 | ALS related data to be stored by TCMS<br>– status data<br>– diagnostic data<br>– configurationinfo | | 2F_03.Zefiro-Europe.CONCEPT.196 |
| 2F_03.Zefiro-Europe.CONCEPT.300 | CBM related data | | 2F_03.Zefiro-Europe.CONCEPT.104 |

TCMS SW to Supplier:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_03.Zefiro-Europe.CONCEPT.33 | Train status data to be provided by TCMS<br>– train speed<br>– Drive direction<br>– logical car number<br>– car orientation<br>– first car of the consist<br>– 400 V AC is present<br>– tbd in ICD document | | 2F_03.Zefiro-Europe.CONCEPT.195 |

# 5References


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[7] DIN EN 15437-2 Railway applications - Axlebox condition monitoring - Performance requirements - Part 2: Onboard systems for temperature monitoring
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